

Helmet Safety Detection with YOLOv10

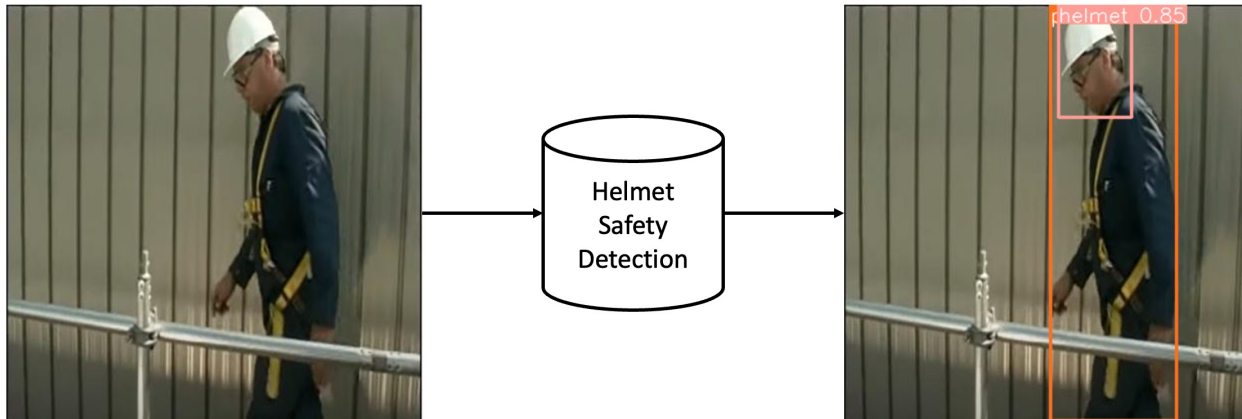
Project



Dinh-Thang Duong – TA

Getting Started

❖ Objectives



```
[root@localhost ~]# ping -q fa.wikipedia.org
PING text.pmtpa.wikimedia.org (208.80.152.2) 56(84) bytes of data:
64 bytes from text.pmtpa.wikimedia.org: icmp_seq=1 ttl=64 time=0.000 ms
--- text.pmtpa.wikimedia.org ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms
[root@localhost ~]# pwd
/root
[root@localhost ~]# cd /var
[root@localhost var]# ls -la
total 72
drwxr-xr-x. 18 root root 4096 Jul 30 22:43 .
drwxr-xr-x. 23 root root 4096 Sep 14 20:42 ..
drwxr-xr-x. 2 root root 4096 May 14 09:15 account
drwxr-xr-x. 11 root root 4096 Jul 31 22:26 cache
drwxr-xr-x. 3 root root 4096 May 18 16:03 db
drwxr-xr-x. 3 root root 4096 May 18 16:03 empty
drwxr-xr-x. 2 root root 4096 May 18 16:03 games
drwxr-xr-x. 2 root gdm 4096 Jun 2 18:39 gdm
drwxr-xr-x. 38 root root 4096 May 18 16:03 lib
drwxr-xr-x. 2 root root 4096 May 18 16:03 local
drwxr-xr-x. 1 root root 11 May 14 09:12 lock -> ../run/lock
drwxr-xr-x. 14 root root 4096 Sep 14 20:42 log
lrwxrwxrwx. 1 root root 19 Jul 30 22:43 mail -> spool/mail
drwxr-xr-x. 2 root root 4096 May 18 16:03 nis
drwxr-xr-x. 2 root root 4096 May 18 16:03 opt
drwxr-xr-x. 2 root root 4096 May 18 16:03 preserve
drwxr-xr-x. 1 root root 11 May 14 09:12 report
lrwxrwxrwx. 1 root root 6 May 14 09:12 run -> ../run
drwxr-xr-x. 14 root root 4096 May 18 16:03 spool
drwxr-xr-x. 4 root root 4096 Sep 12 23:50 tmp
drwxr-xr-x. 2 root root 4096 May 18 16:03 yp
[root@localhost var]# yum search wiki
loaded plugins: langpacks, presto, refresh-packagekit, remove-with-leaves
rpfusion-free-updates                               2.7 kB  00:00
rpfusion-free-updates/primary_db                    206 kB  00:04
rpfusion-nonfree-updates                             2.7 kB  00:00
updates/metalink                                    5.9 kB  00:00
updates                                              4.7 kB  00:00
updates/primary_db                                73% [=====] 1 62 kB/s | 2.6 MB 00:15 ETA
```



Our objectives:

- Discuss about one of famous AI application in computer vision: Object Detection.
- Learn how to use and train YOLOv10 model on downstream task.
- Learn about Linux commands to interact in terminal.
- Learn how to label new images.

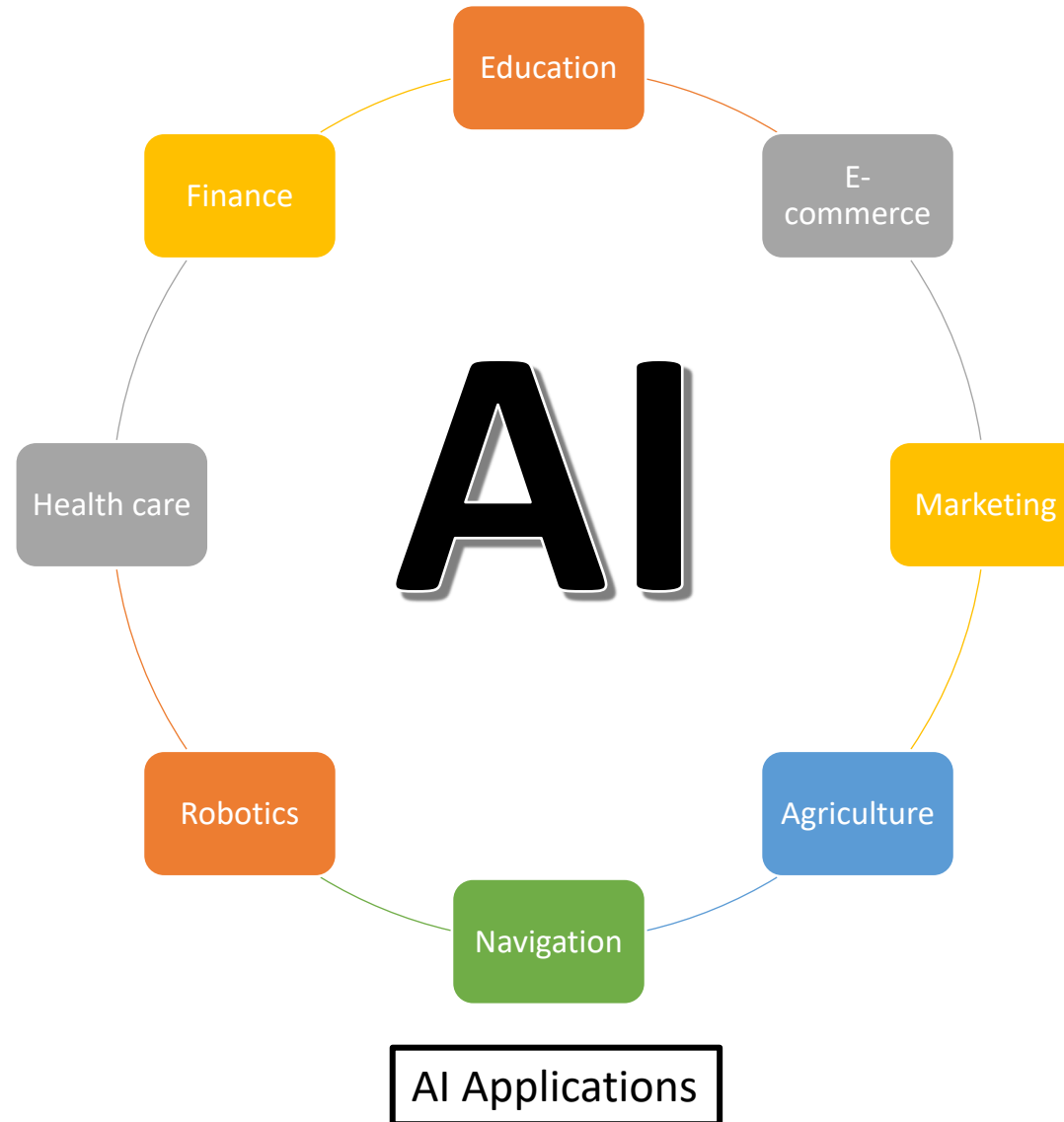
Outline

- Introduction
- Code Environment
- YOLOv10
- Linux Commands
- LabelImg
- Question

Introduction

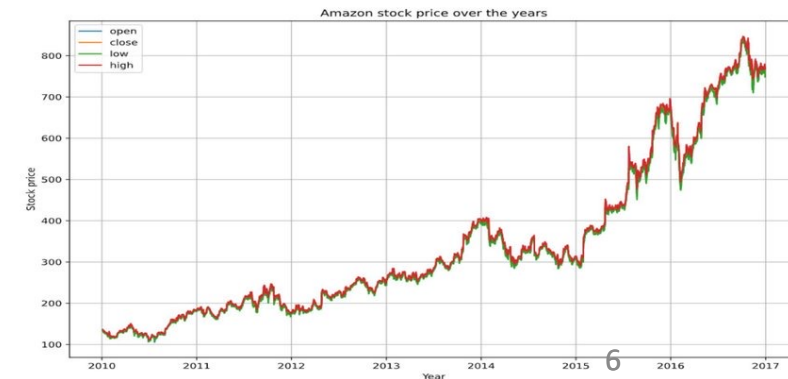
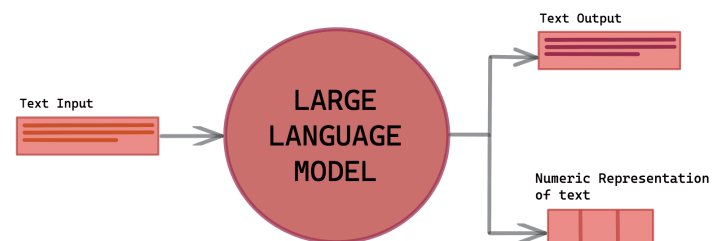
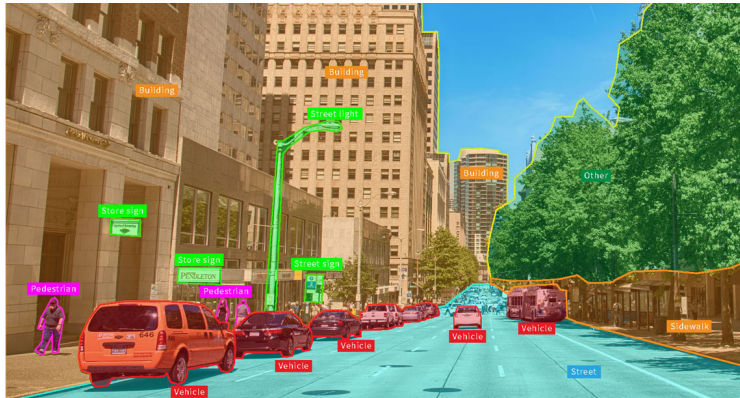
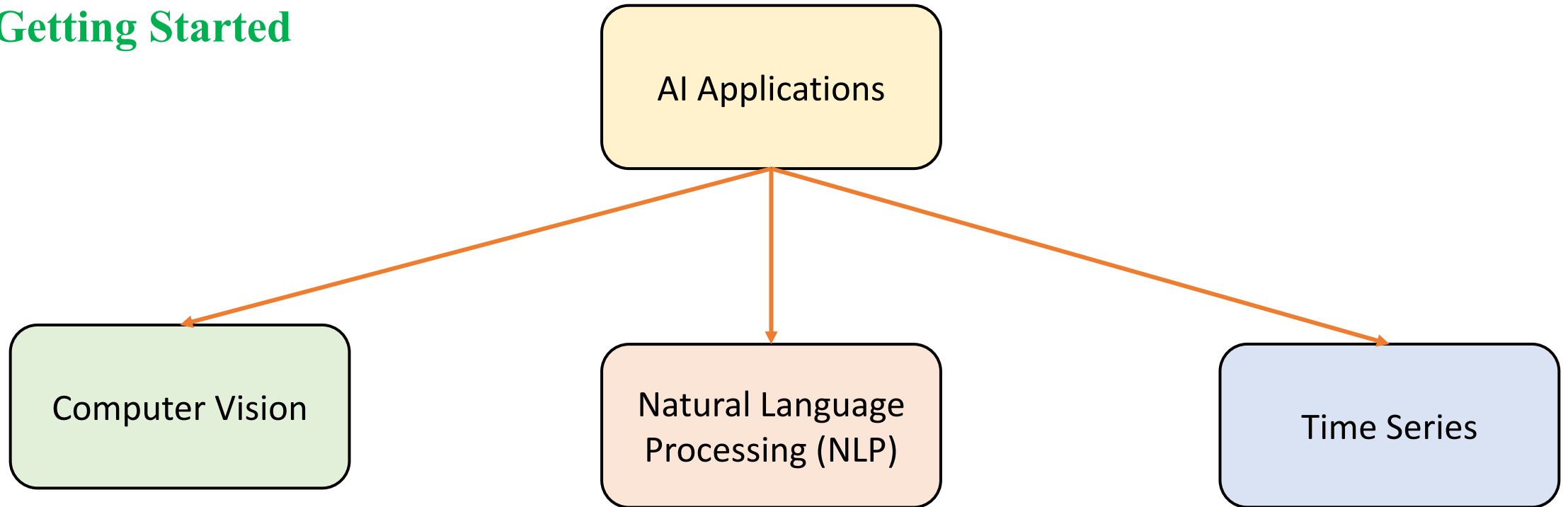
Introduction

❖ Getting Started



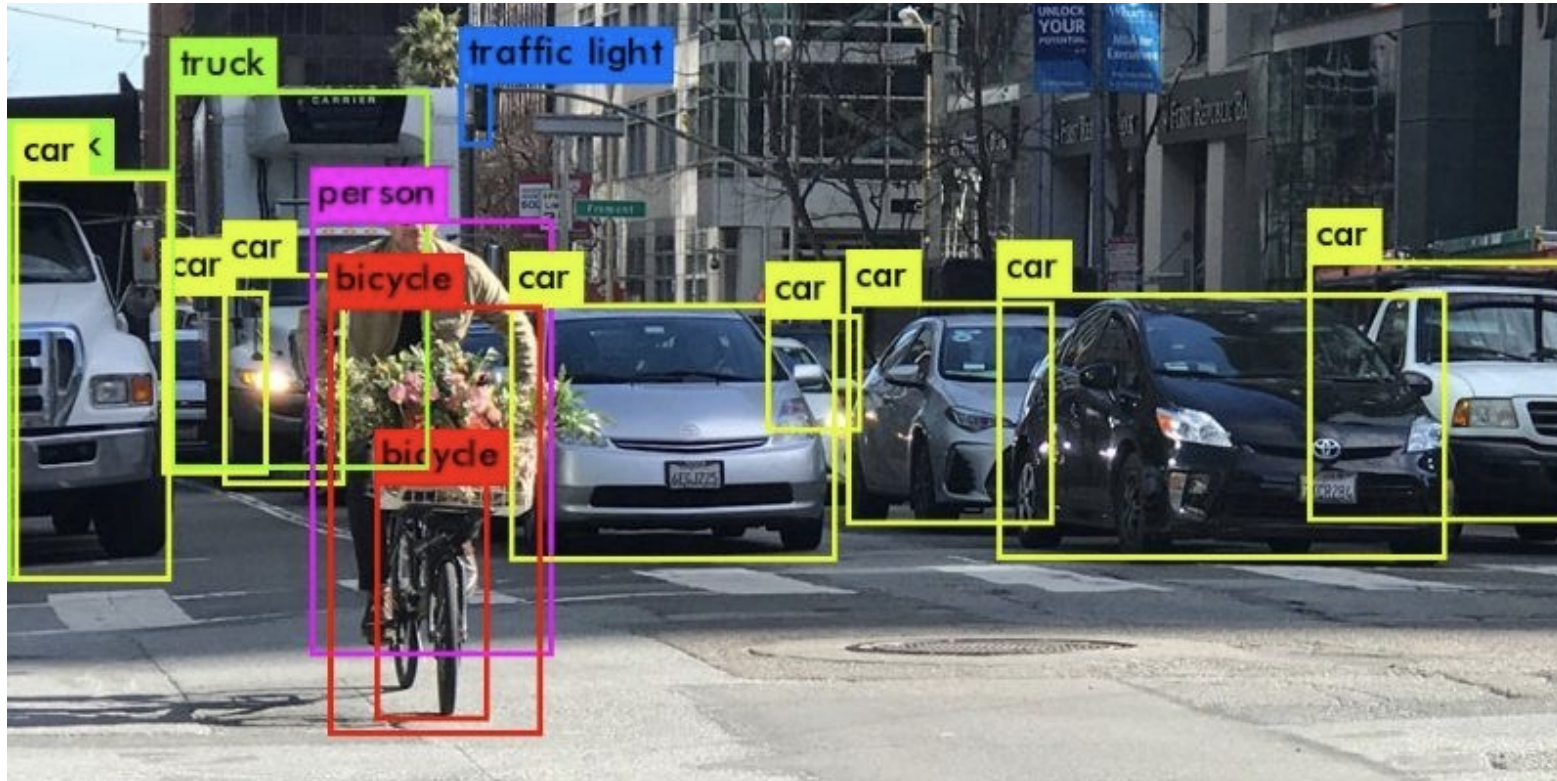
Introduction

❖ Getting Started



Introduction

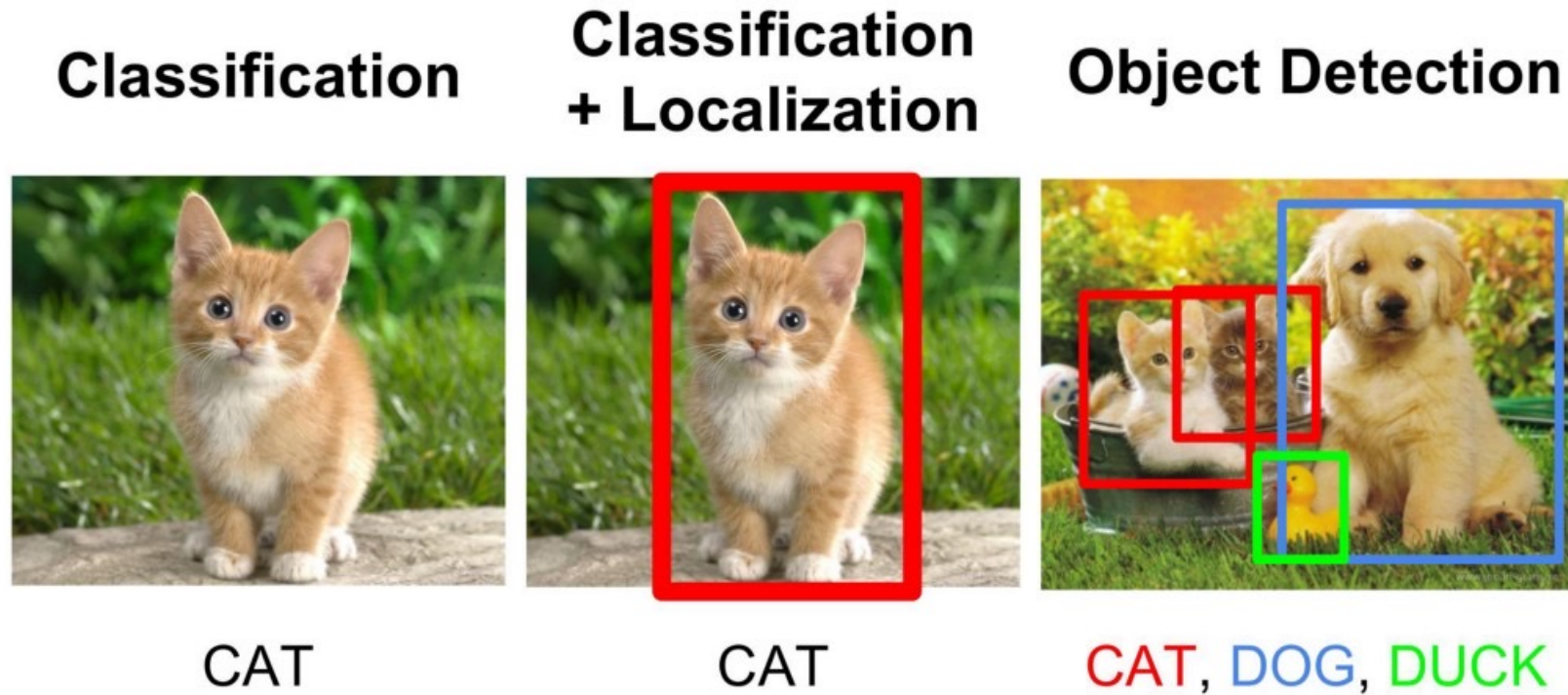
❖ Getting Started



Object Detection: a Computer Vision task for identifying objects in an image.

Introduction

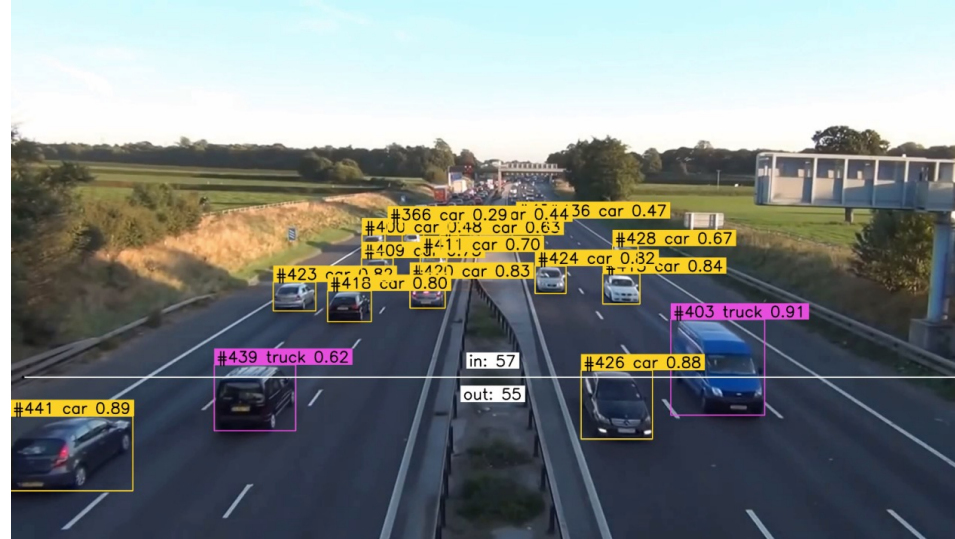
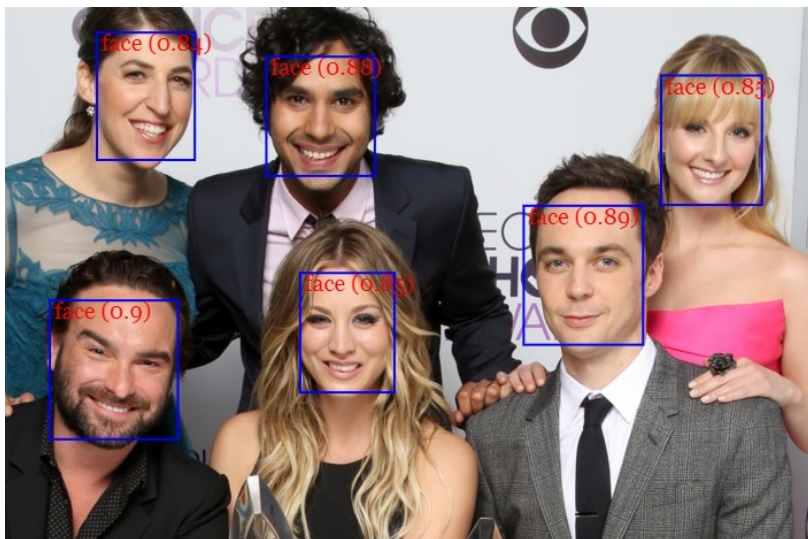
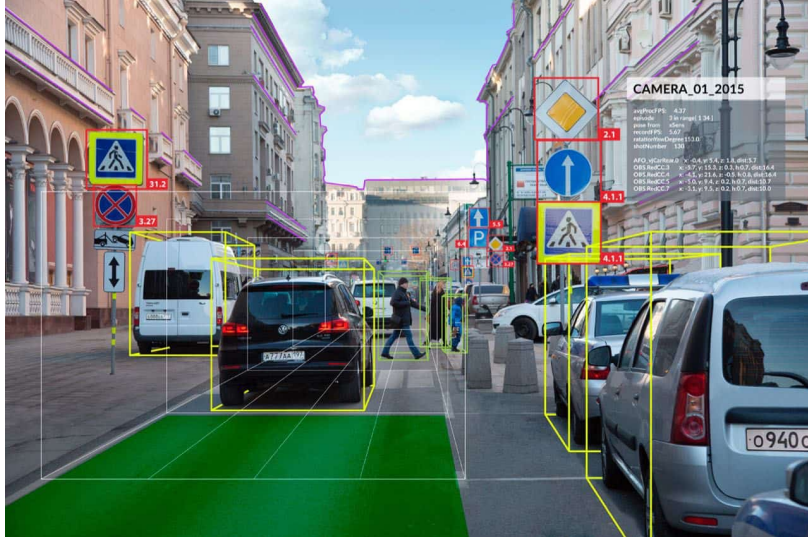
❖ What is Object Detection?



Object Detection = Classification + Localization

Introduction

❖ Object Detection Applications



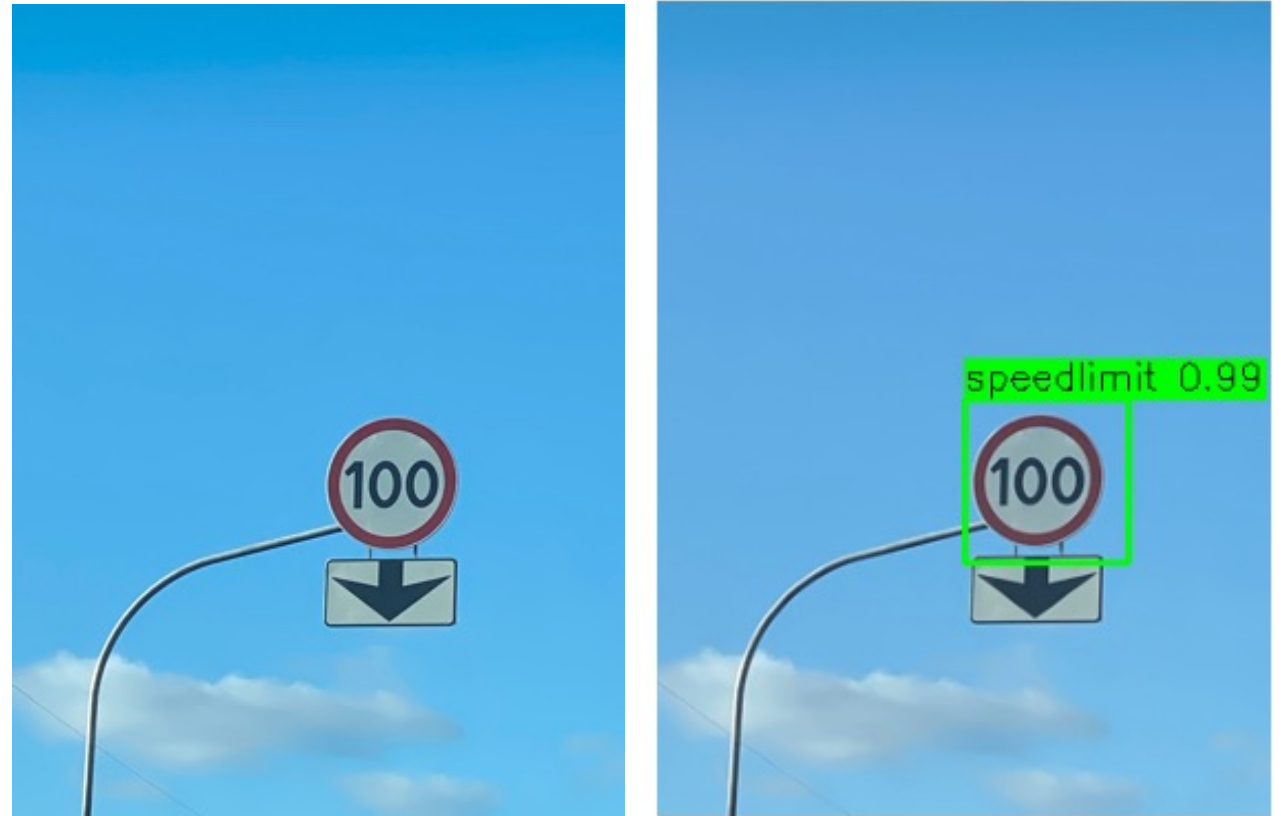
Some Object
Detection
Applications

Introduction

❖ Object Detection Input/Output

Input: An image.

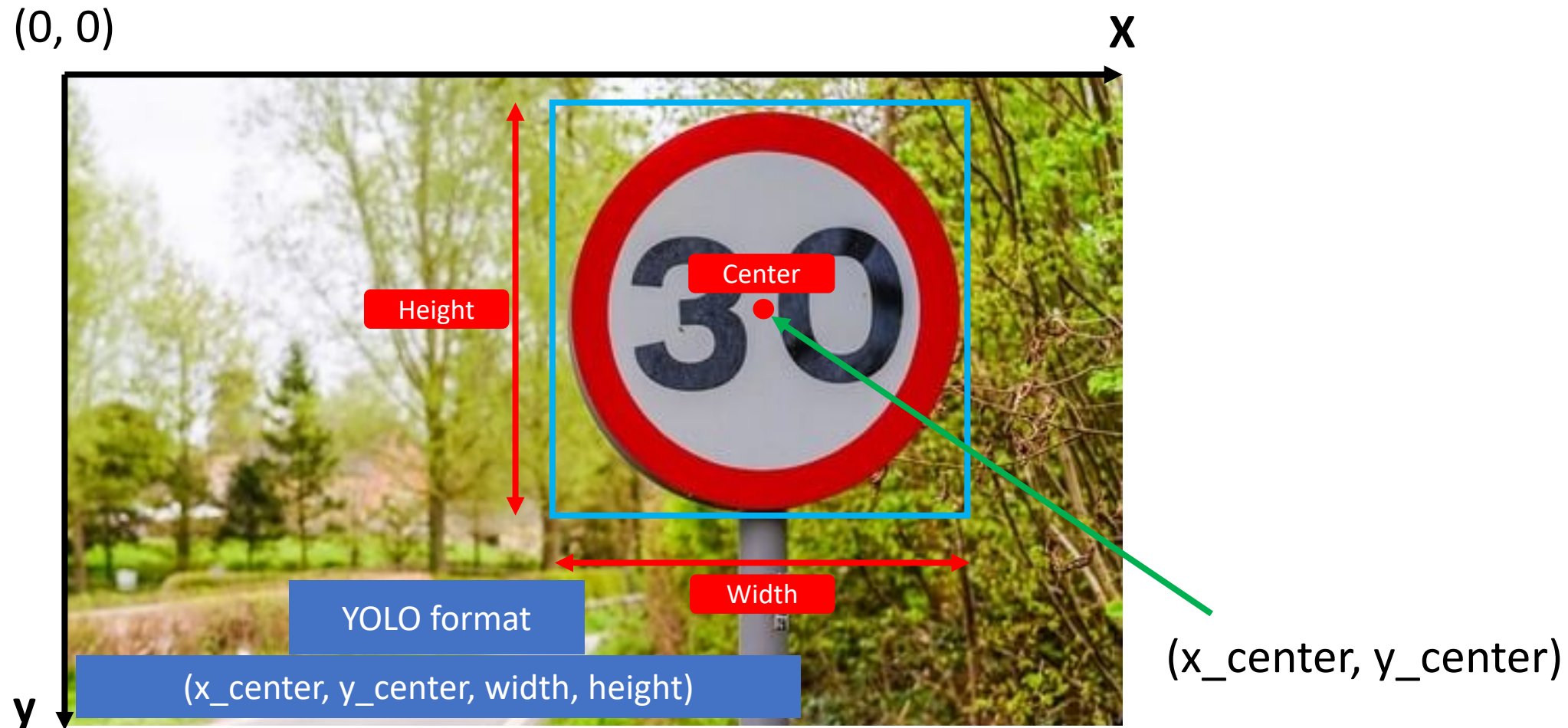
Output: A list of predicted bounding boxes in the form of (xmin, ymin, xmax, ymax, confidence_score)



Output Visualization

Introduction

❖ Object Localization: Bounding box



Introduction

❖ Object Detection Confidence Score

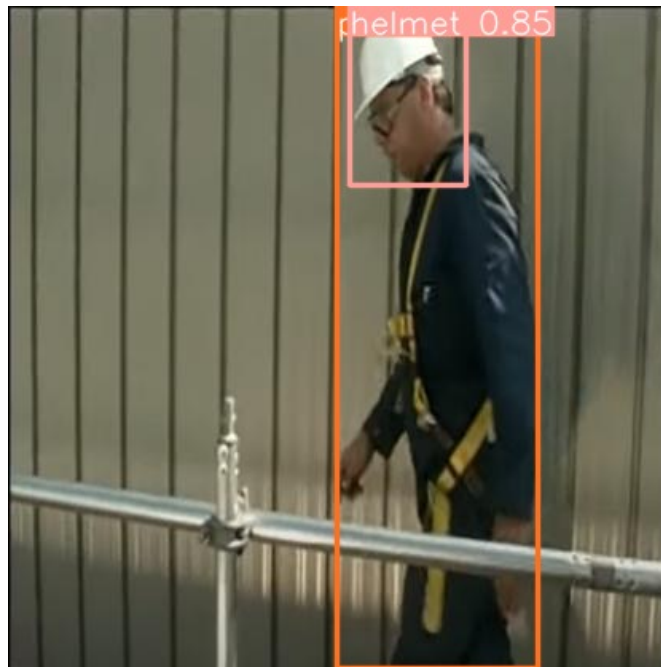


Confidence Score: A value represents how 'confident' the model is to assign a class to the corresponding object.

Introduction

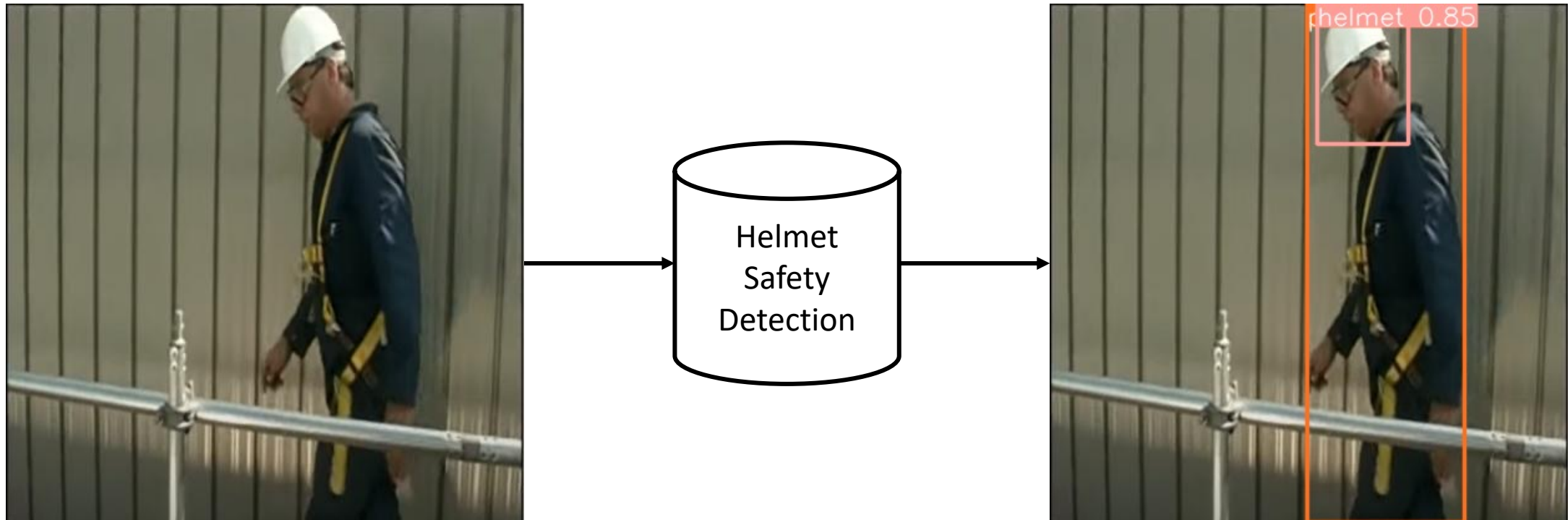
❖ Project Description

Problem statement: Given a Helmet Safety Dataset, train a YOLOv10 model to detect whether people in an image are wearing a helmet or not?



Introduction

❖ Helmet Safety Detection Problem



Code Environment

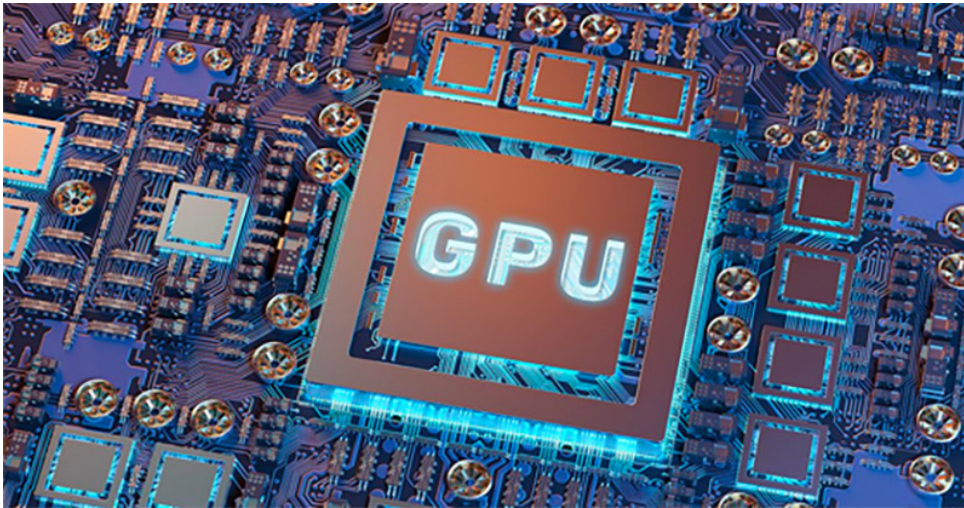
❖ Google Colab Introduction



Google Colaboratory: A data analysis and machine learning tool that allows us to combine executable Python code and rich text.

Code Environment

❖ Google Colab Advantages



- Free GPU.
- Pre-installed popular ML packages.
- Notebook interface.

```
1 !pip list
```

```
tensorflow                2.12.0
tensorflow-datasets       4.9.2
tensorflow-estimator      2.12.0
tensorflow-gcs-config     2.12.0
tensorflow-hub            0.13.0
tensorflow-io-gcs-filesystem 0.32.0
tensorflow-metadata       1.13.1
tensorflow-probability    0.20.1
tensorstore               0.1.36
termcolor                 2.3.0
terminado                 0.17.1
text-unidecode            1.3
textblob                  0.17.1
tf-slim                   1.1.0
thinc                     8.1.9
threadpoolctl             3.1.0
tifffile                  2023.4.12
tinycss2                  1.2.1
toml                       0.10.2
tomli                     2.0.1
toolz                     0.12.0
torch                     2.0.1+cu118
torchaudio                2.0.2+cu118
torchdata                  0.6.1
torchsummary              1.5.1
torchtext                 0.15.2
torchvision               0.15.2+cu118
```


Code Environment

❖ Google Colab Instruction

Welcome To Colaboratory

File Edit View Insert Runtime Tools Help

Table of contents

- Getting started
- Data science
- Machine learning
- More Resources
- Featured examples
- Section

+ Code + Text Copy to Drive

Share Sign in

Welcome to Colab!

If you're already familiar with Colab, check out this video to learn about interactive tables, the executed code history view, and the command palette.

3 Cool Google Colab Features

Sign in with Google Account

What is Colab?

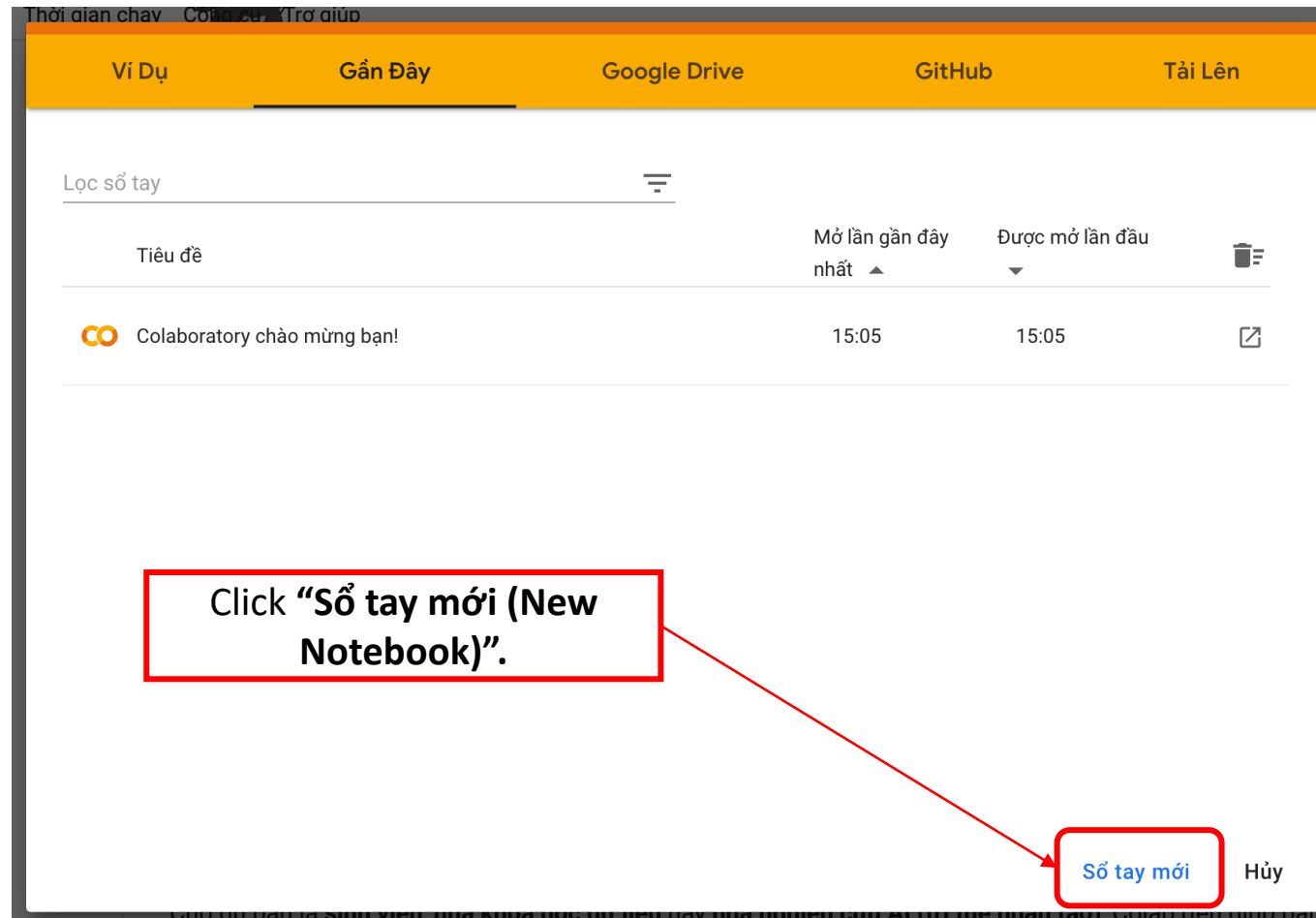
Colab, or "Colaboratory", allows you to write and execute Python in your browser, with

- Zero configuration required
- Access to GPUs free of charge
- Easy sharing

Whether you're a student, a data scientist or an AI researcher, Colab can make your work easier. Watch [Introduction to Colab](#) to learn more or

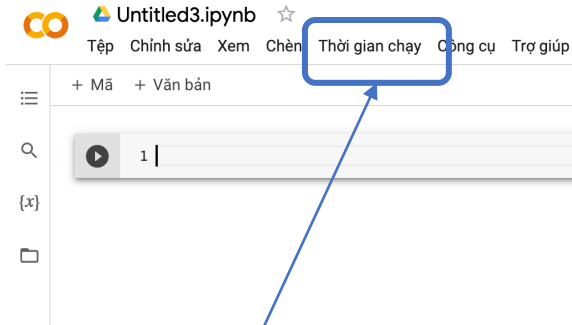
Code Environment

❖ Google Colab Instruction

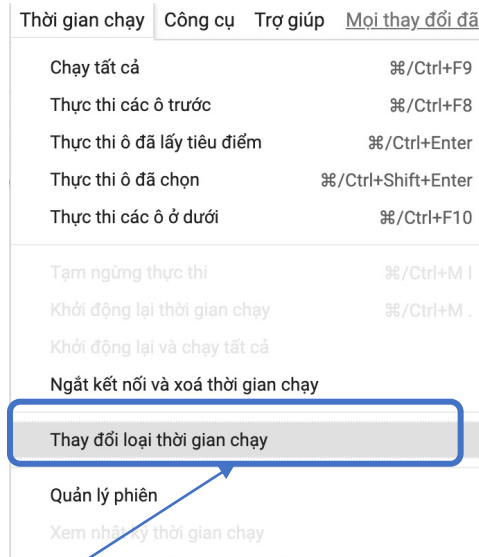


Code Environment

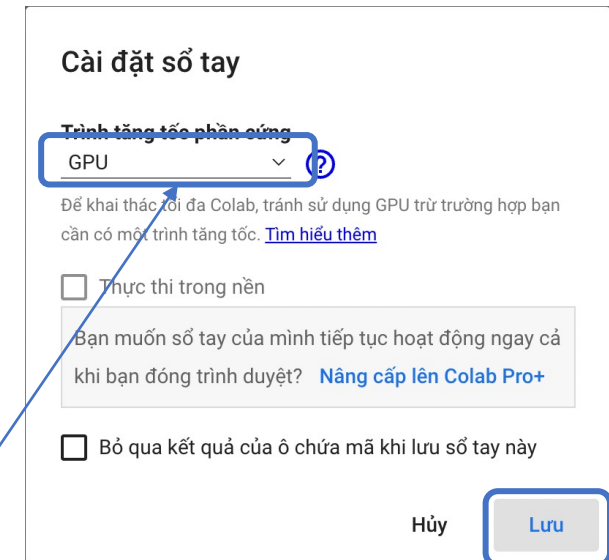
❖ Google Colab Instruction



1. Click “Thời gian chạy (Runtime)”.



2. Click “Thay đổi loại thời gian chạy (Change Runtime Type)”.



3. Click to change from **None** to **GPU**.

4. Click “Lưu (Save)”.

Code Environment

❖ Google Colab Instruction

The screenshot shows the Google Colab interface for an 'Untitled3.ipynb' file. The top navigation bar includes options like 'Tệp', 'Chỉnh sửa', 'Xem', 'Chèn', 'Thời gian chạy', 'Công cụ', 'Trợ giúp', and 'Mọi thay đổi đã được lưu'. On the right, there are buttons for 'Nhận xét', 'Chia sẻ', and 'Kết nối'. The 'Kết nối' button is highlighted with a red box. A red arrow points from a text box to this button. Below the 'Kết nối' button is a dropdown menu with the text 'Nhập để kết nối'. A green arrow points from another text box to the status bar at the bottom, which shows a green checkmark, 'RAM', and 'Ổ đĩa'. At the very bottom, a dark grey box displays the connection status: 'Đã kết nối với Python 3 Phần phụ trợ Google Compute Engine (GPU)' and resource usage: 'RAM: 0.78 GB/12.68 GB Ổ đĩa: 38.73 GB/78.19 GB'.

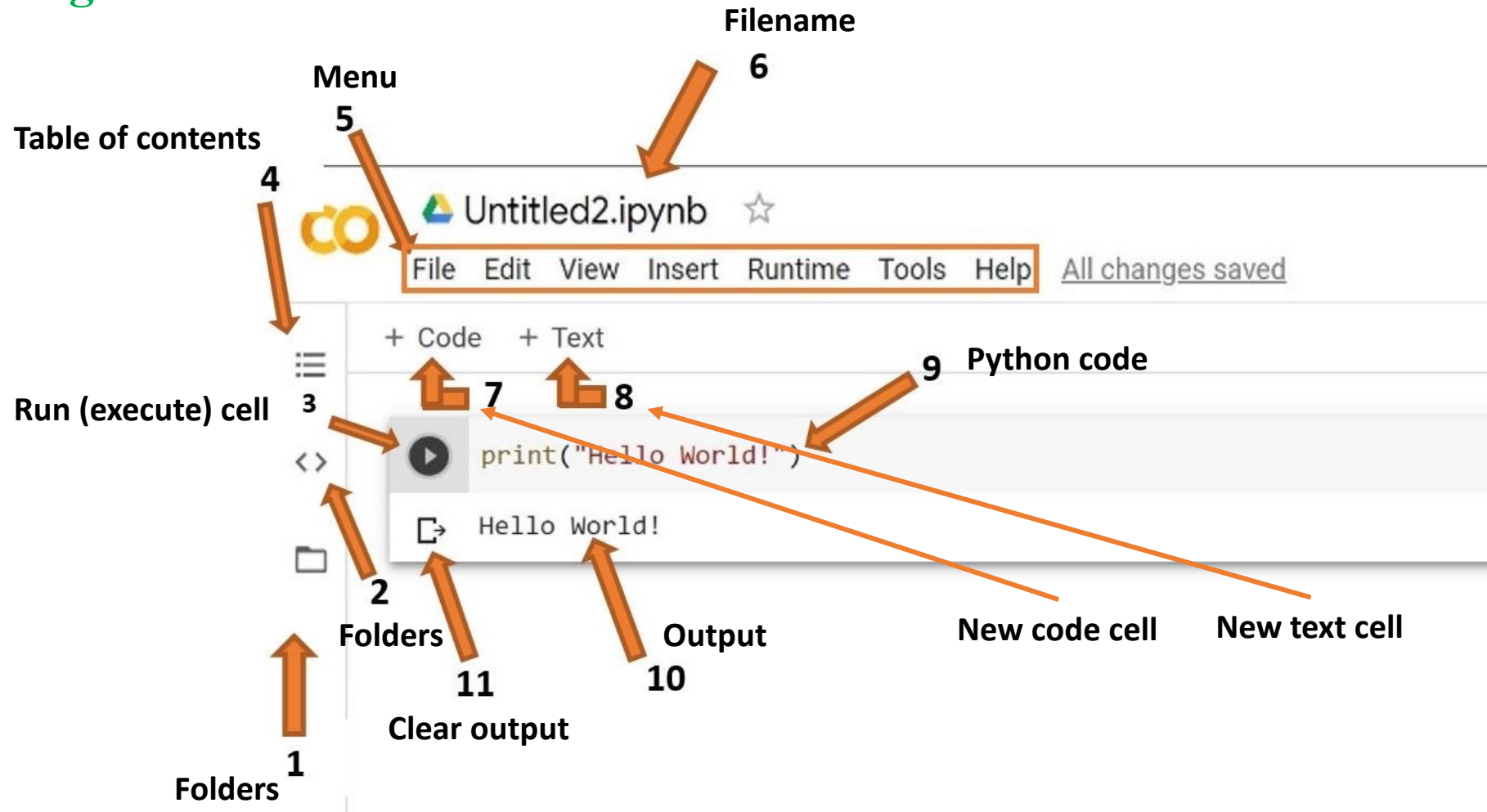
1. Click “Kết nối (Connect)”.

2. Wait until green check mark appear.

Đã kết nối với Python 3 Phần phụ trợ Google Compute Engine (GPU)
RAM: 0.78 GB/12.68 GB Ổ đĩa: 38.73 GB/78.19 GB

Code Environment

❖ Google Colab Interface



YOLOv10

❖ Introduction

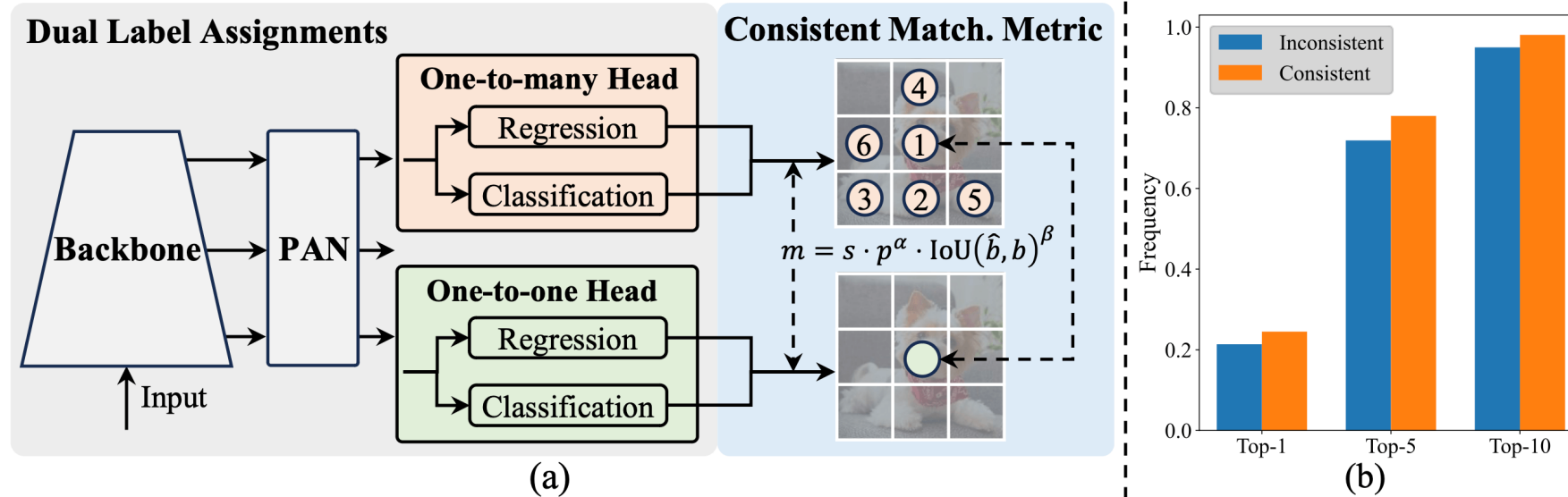
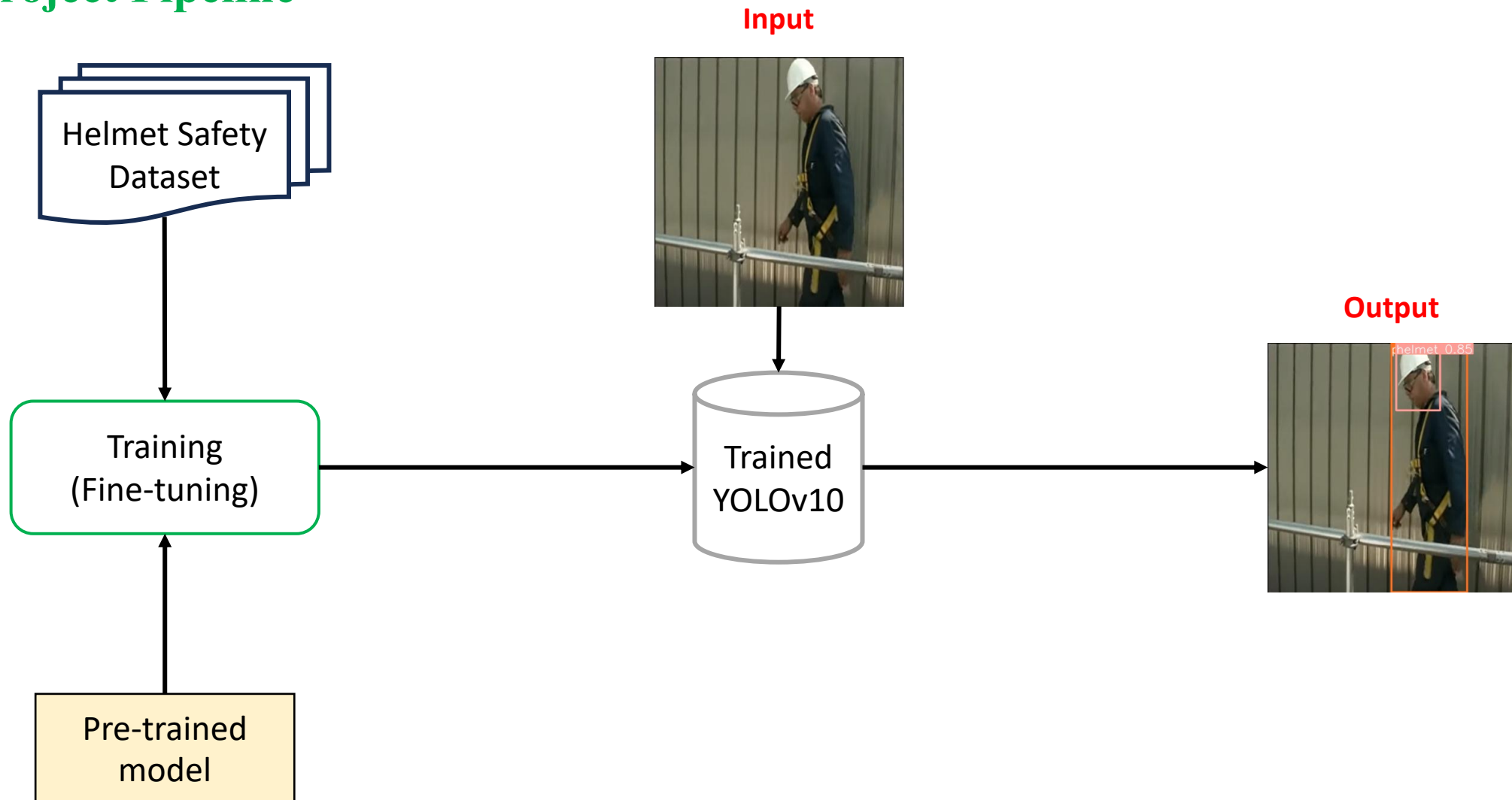


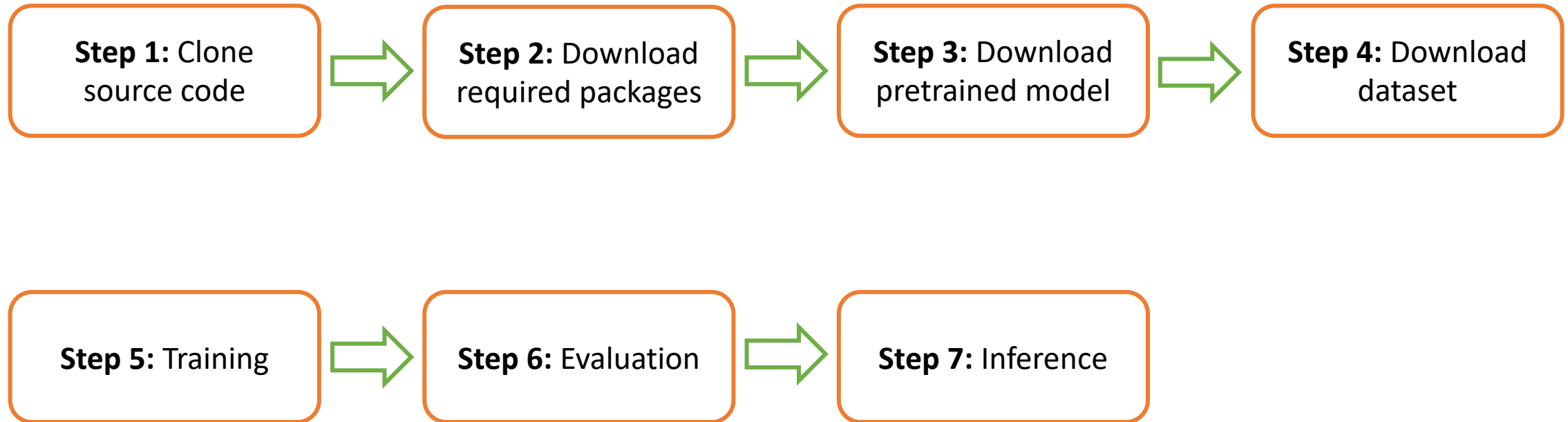
Figure 2: (a) Consistent dual assignments for NMS-free training. (b) Frequency of one-to-one assignments in Top-1/5/10 of one-to-many results for YOLOv8-S which employs $\alpha_{o2m}=0.5$ and $\beta_{o2m}=6$ by default [20]. For consistency, $\alpha_{o2o}=0.5$; $\beta_{o2o}=6$. For inconsistency, $\alpha_{o2o}=0.5$; $\beta_{o2o}=2$.

YOLOv10: A family of object detection architectures and models pretrained on the COCO dataset

❖ Project Pipeline



❖ YOLOv10 Installation and Usage



YOLOv10

❖ Step 1: Clone YOLOv10 source code

The screenshot shows the GitHub repository for YOLOv10. The repository is public and has 39 watchers, 582 forks, and 7.6k stars. The main branch is selected. The repository contains several folders and files, including docker, docs, examples, figures, logs, tests, ultralytics, .gitignore, .pre-commit-config.yaml, CONTRIBUTING.md, LICENSE, README.md, and app.py. The right sidebar shows the repository's description: "YOLOv10: Real-Time End-to-End Object Detection", a link to the arXiv paper, and a list of releases, including the latest one: "Update checkpoints with oth...".

File/Folder	Description	Last Commit
leonnil	ddp resume checkpoint fix (#184)	2c36ab0 · last week
docker	Optimize TFJS export on ARM64 (#8946)	3 months ago
docs	docs: update view-results-in-terminal.md (#66)	last month
examples	Update Discord and Contributing Guide URLs (#9270)	3 months ago
figures	add yolov10	last month
logs	add logs	3 weeks ago
tests	Update pyproject.toml [tool.pytest.ini_options] (#925...)	3 months ago
ultralytics	ddp resume checkpoint fix (#184)	last week
.gitignore	ultralytics 8.1.26 LoadImagesAndVideos batched infere...	3 months ago
.pre-commit-config.yaml	YAML reformat (#7669)	5 months ago
CONTRIBUTING.md	ultralytics 8.0.224 Counting and Heatmaps updates (#...	6 months ago
LICENSE	Update LICENSE to AGPL-3.0 (#2031)	last year
README.md	update README	last week
app.py	Added video output support for gradio (#113)	3 weeks ago

YOLOv10 GitHub Repository

source: <https://github.com/THU-MIG/yolov10>

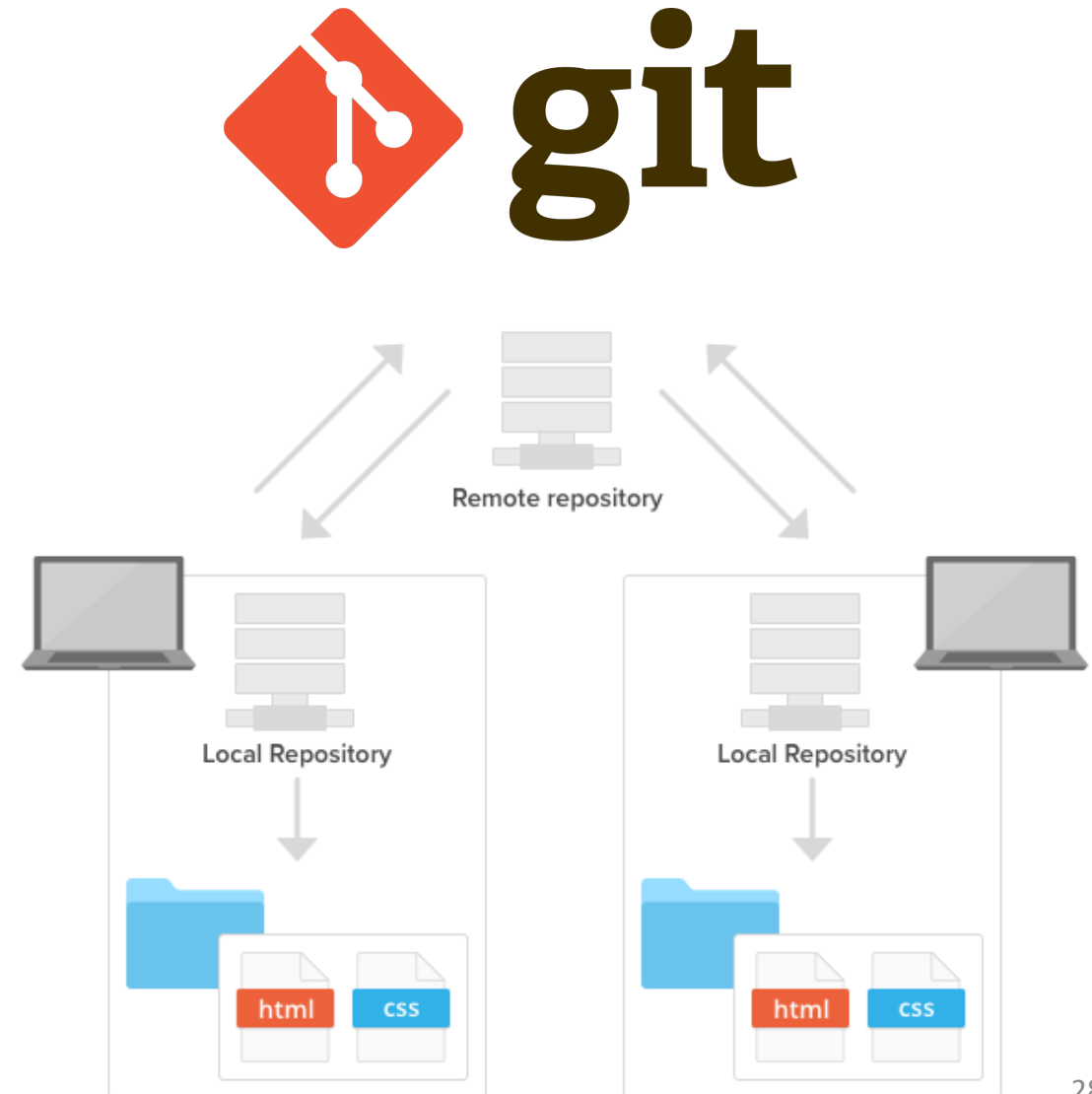
❖ Step 1.1: About GitHub



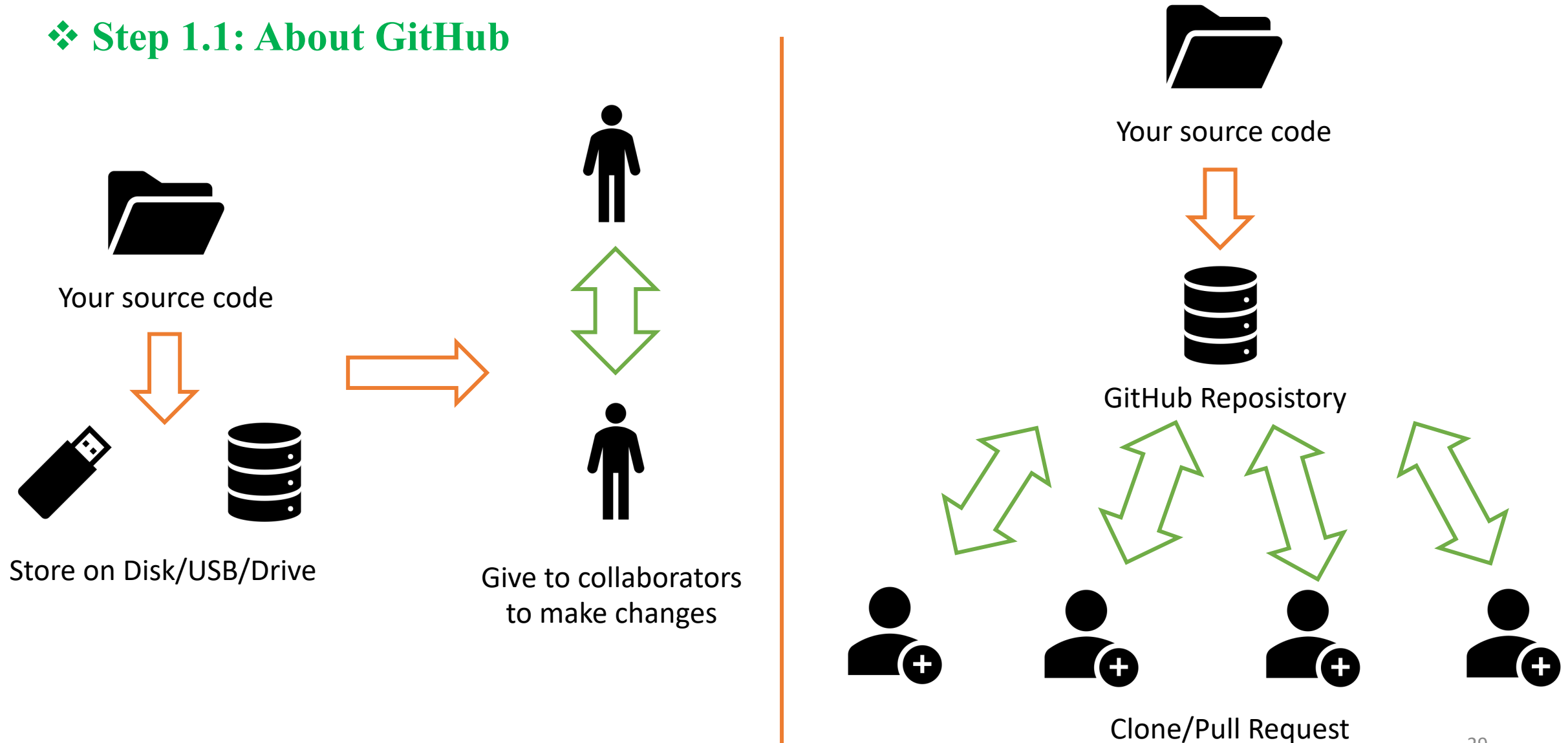
A code hosting platform for collaboration and version control

GitHub usages:

1. Store your code
2. Work together
3. Find source code from other people



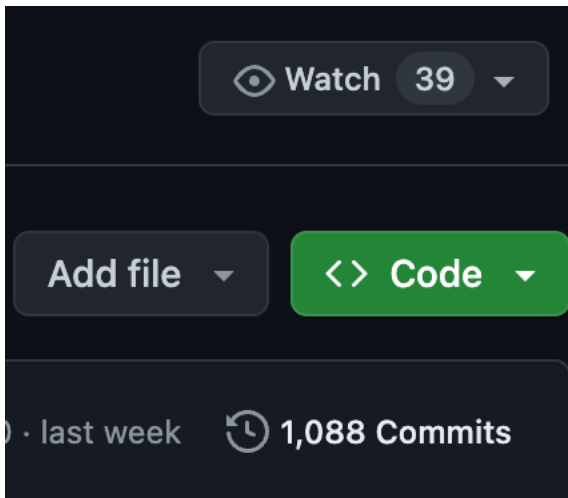
❖ Step 1.1: About GitHub



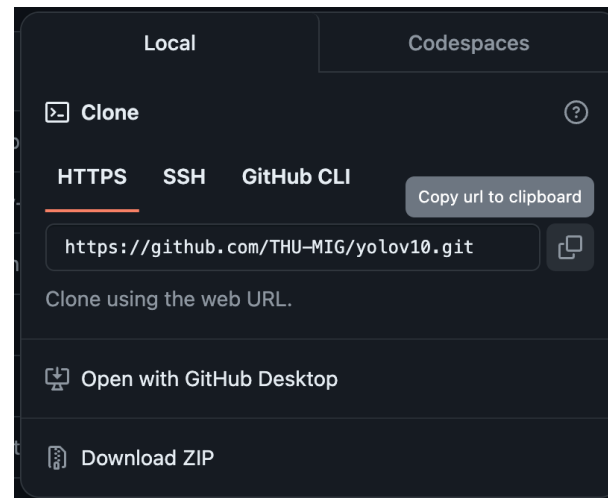
YOLOv10

❖ Step 1: Clone YOLOv8 source code

Step 1: Click Code



Step 2: Copy link



Step 3: Clone source code

```
1 !git clone https://github.com/THU-MIG/yolov10.git
```

```
Cloning into 'yolov10'...
remote: Enumerating objects: 20299, done.
remote: Counting objects: 100% (1342/1342), done.
remote: Compressing objects: 100% (122/122), done.
remote: Total 20299 (delta 1299), reused 1220 (delta 1220), pack-reused 18957
Receiving objects: 100% (20299/20299), 11.18 MiB | 12.82 MiB/s, done.
Resolving deltas: 100% (14302/14302), done.
```

*Clone source code command:

`git clone <link>`: Clone (copy) a repository into a new directory.

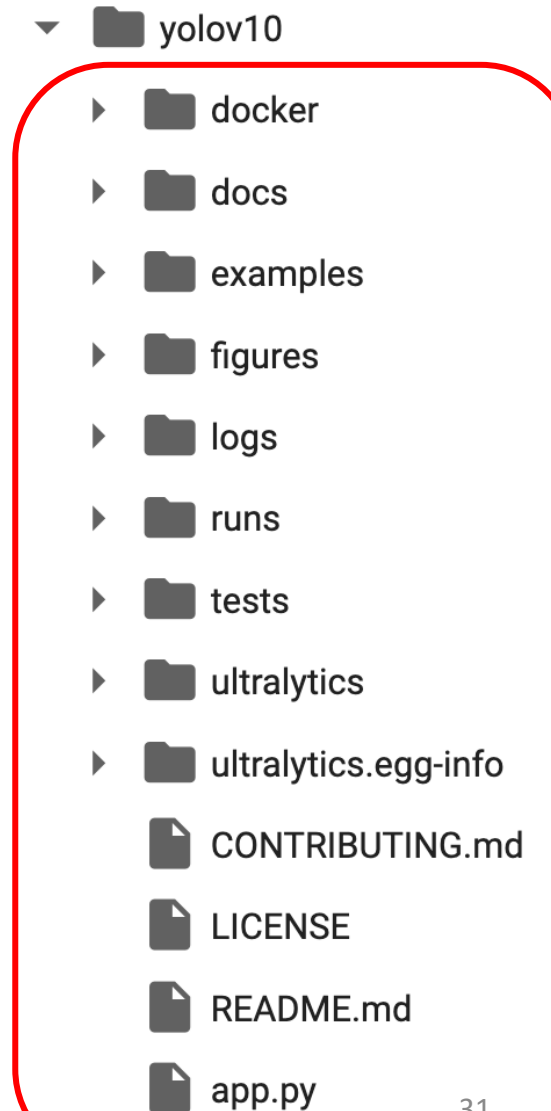
Remote
repository

Local

YOLOv10

❖ Step 1: Clone YOLOv10 source code

folder	docker	Optimize TFJS export on ARM64 (#8946)	3 months ago
folder	docs	docs: update view-results-in-terminal.md (#66)	last month
folder	examples	Update Discord and Contributing Guide URLs (#9270)	3 months ago
folder	figures	add yolov10	last month
folder	logs	add logs	3 weeks ago
folder	tests	Update pyproject.toml [tool.pytest.ini_options] (#925...	3 months ago
folder	ultralytics	ddp resume checkpoint fix (#184)	last week
file	.gitignore	ultralytics 8.1.26 LoadImagesAndVideos batched infer...	3 months ago
file	.pre-commit-config.yaml	YAML reformat (#7669)	5 months ago
file	CONTRIBUTING.md	ultralytics 8.0.224 Counting and Heatmaps updates (#...	6 months ago
file	LICENSE	Update LICENSE to AGPL-3.0 (#2031)	last year
file	README.md	update README	last week
file	app.py	Added video output support for gradio (#113)	3 weeks ago



❖ Step 2: Download required packages

- ▼ yolo10
 - docker
 - docs
 - examples
 - figures
 - logs
 - runs
 - tests
 - ultralytics
 - ultralytics.egg-info
 - CONTRIBUTING.md
 - LICENSE
 - README.md
 - app.py
 - flops.py
 - mkdocs.yml
 - pyproject.toml
 - requirements.txt**
 - yolo10n.pt
 - yolo8n.pt



requirements.txt ✕

```
1 torch==2.0.1
2 torchvision==0.15.2
3 onnx==1.14.0
4 onnxruntime==1.15.1
5 pycocotools==2.0.7
6 PyYAML==6.0.1
7 scipy==1.13.0
8 onnxsim==0.4.36
9 onnxruntime-gpu==1.18.0
10 gradio==4.31.5
11 opencv-python==4.9.0.80
12 psutil==5.9.8
13 py-cpuinfo==9.0.0
14 huggingface-hub==0.23.2
15 safetensors==0.4.3
```

Download all
these packages to
be able to run
YOLOv10

❖ Step 2: Download required packages

- ▼ yolo10
 - ▶ docker
 - ▶ docs
 - ▶ examples
 - ▶ figures
 - ▶ logs
 - ▶ runs
 - ▶ tests
 - ▶ ultralytics
 - ▶ ultralytics.egg-info
 - CONTRIBUTING.md
 - LICENSE
 - README.md
 - app.py
 - flops.py
 - mkdocs.yml
 - pyproject.toml
 - requirements.txt
 - yolo10n.pt
 - yolo10n.pt



pyproject.toml X

```
64
65 # Required dependencies -----
66 dependencies = [
67     "matplotlib>=3.3.0",
68     "opencv-python>=4.6.0",
69     "pillow>=7.1.2",
70     "pyyaml>=5.3.1",
71     "requests>=2.23.0",
72     "scipy>=1.4.1",
73     "torch>=1.8.0",
74     "torchvision>=0.9.0",
75     "tqdm>=4.64.0", # progress bars
76     "psutil", # system utilization
77     "py-cpuinfo", # display CPU info
78     "thop>=0.1.1", # FLOPs computation
79     "pandas>=1.1.4",
80     "seaborn>=0.11.0", # plotting
81 ]
```

Download all
these packages to
be able to run
YOLOv10

YOLOv10

❖ Step 2: Download required packages

Step 1: Move to YOLOv10

Files

📁 📁 📁 🔍 You are here

📁 ..
📁 sample_data
📁 ultralytics

1 !pwd
/content

↓

📁 sample_data
📁 yolov10

1 %cd yolov10
/content/yolov10

Move to here

Step 2: Install packages

```
1 !pip install -e .
```

Looking in indexes: <https://pypi.org/simple>, <https://us-python.pkg.dev/colab-wheels/public/simple/>
Obtaining file:///content/ultralytics
Preparing metadata (setup.py) ... done
Requirement already satisfied: matplotlib>=3.2.2 in /usr/local/lib/python3.10/dist-packages (from ultralytics)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)
Requirement already satisfied: Pillow>=7.1.2 in /usr/local/lib/python3.10/dist-packages (from ultralytics)
Requirement already satisfied: PyYAML>=5.3.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics)
Requirement already satisfied: torch>=1.7.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)
Requirement already satisfied: torchvision>=0.8.1 in /usr/local/lib/python3.10/dist-packages (from ultralytics)
Requirement already satisfied: tqdm>=4.64.0 in /usr/local/lib/python3.10/dist-packages (from ultralytics)
Requirement already satisfied: pandas>=1.1.4 in /usr/local/lib/python3.10/dist-packages (from ultralytics)

Full commands:

```
1 %cd yolov10  
2 !pip install -q -r requirements.txt  
3 !pip install -e .
```

YOLOv10

❖ Step 3: Download pretrained models

Model	Test Size	#Params	FLOPs	AP ^{val}	Latency
YOLOv10-N	640	2.3M	6.7G	38.5%	1.84ms
YOLOv10-S	640	7.2M	21.6G	46.3%	2.49ms
YOLOv10-M	640	15.4M	59.1G	51.1%	4.74ms
YOLOv10-B	640	19.1M	92.0G	52.5%	5.74ms
YOLOv10-L	640	24.4M	120.3G	53.2%	7.28ms
YOLOv10-X	640	29.5M	160.4G	54.4%	10.70ms

Pretrained models: models that have been trained on a large dataset

❖ Step 3: Download pretrained models

Step 1: Right click and copy link address

Model	Test Size	#Params	FLOPs	AP ^{val}	Latency
YOLOv10-N	640	2.3M	6.7G	38.5%	1.84ms
YOLOv10-S	640	7.2M	21.6G	46.3%	2.49ms
YOLOv10-M	640	15.4M	59.1G	51.1%	4.74ms
YOLOv10-B	640	19.1M	92.0G	52.5%	5.74ms
YOLOv10-L	640	24.4M	120.3G	53.2%	7.28ms
YOLOv10-X	640	29.5M	160.4G	54.4%	10.70ms

source: <https://github.com/THU-MIG/yolov10>

Step 2: Download file using wget command

```
1 # Nano version: yolov10n.pt
2 !wget https://github.com/THU-MIG/yolov10/releases/download/v1.1/yolov10n.pt
```

```
--2024-06-21 13:43:27-- https://github.com/THU-MIG/yolov10/releases/download/v1.1/yolov10n.pt
Resolving github.com (github.com)... 140.82.112.3
Connecting to github.com (github.com)|140.82.112.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
```

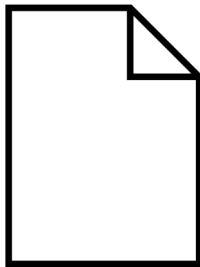
Command: `wget <link>`

- CONTRIBUTING.md
- LICENSE
- README.md
- app.py
- flops.py
- mkdocs.yml
- pyproject.toml
- requirements.txt
- yolov10n.pt**

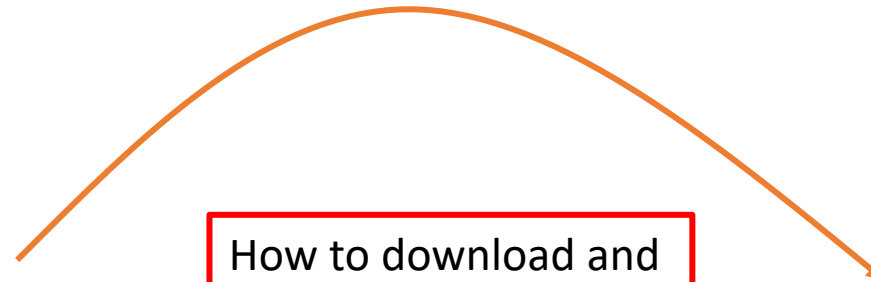
When execution complete,
pretrained file will be placed
in ultralytics folder.

❖ Step 4: Download dataset

- Given a Car Detection Dataset in YOLO data structure format.
- URL is Google Drive: <https://drive.google.com/file/d/1sX9qmlPY1SXwboz9qOzjyLrkzCwYoT31/view?usp=sharing>



Dataset file

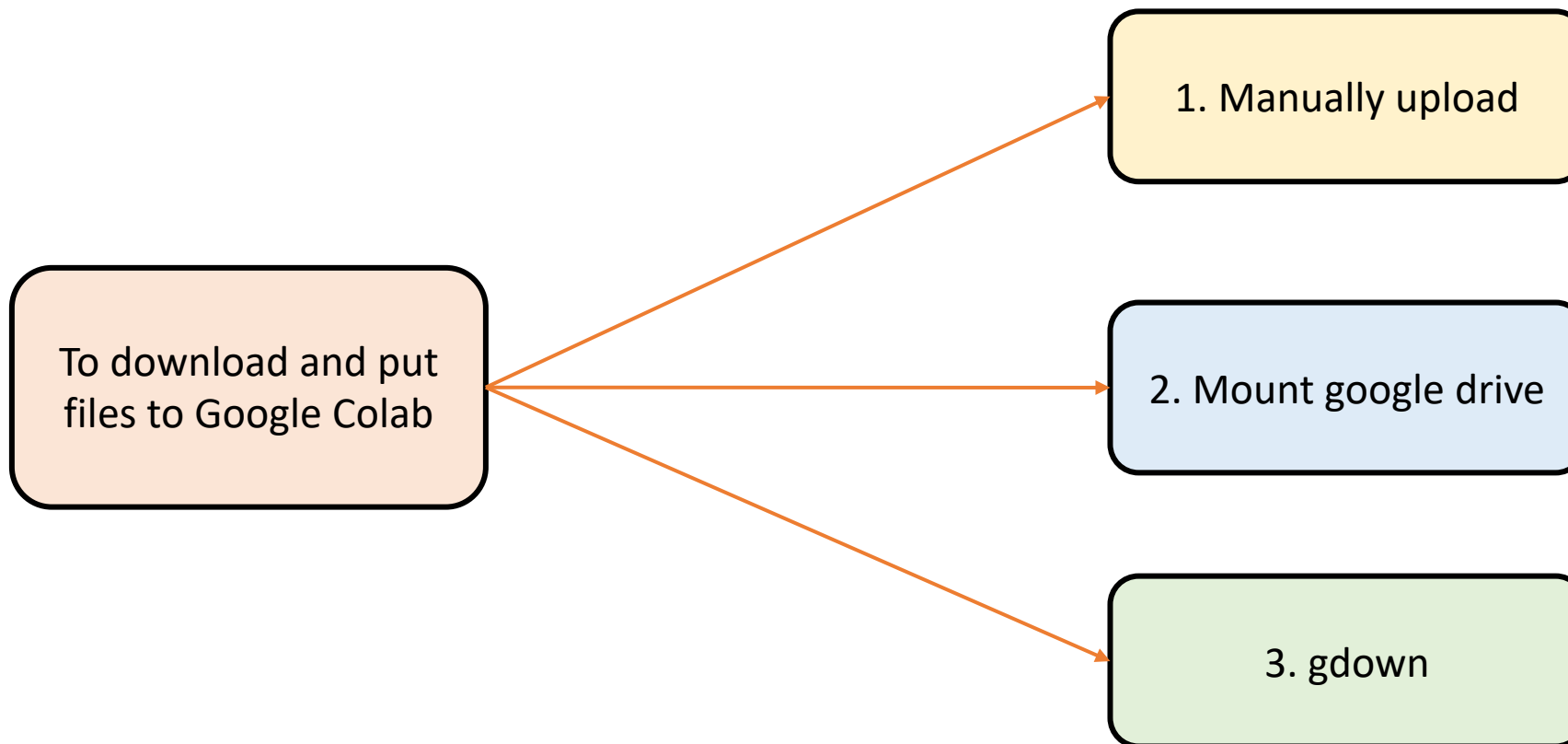


How to download and
put dataset file to
Google Colab



❖ Step 4: Download dataset

- Given a Helmet Safety Dataset in YOLO data structure format.
- URL is Google Drive: <https://drive.google.com/file/d/1twdtZEfcw4ghSZliPDypJurZnNXzMO7R/view?usp=sharing>



❖ Step 4: Download dataset

Image:



Label:

helmet-246_.jpg.rf.8fd98ab9dbccf6b602216cd35c132fac.txt X

```
1 1 0.35336538461538464 0.36658653846153844 0.125 0.24759615384615385
2 1 0.59375 0.3004807692307692 0.17307692307692307 0.24519230769230768
```

YOLOv10 Dataset folder format

▼ safety_helmet_dataset

▼ test

▶ images

▶ labels

▼ train

▶ images

▶ labels

▼ valid

▶ images

▶ labels

📄 frame007.25.00-07.30.00.jpg

📄 frame007.30.00-07.35.00.jpg

📄 frame007.35.00-07.40.00.jpg

📄 frame007.40.00-07.45.00.jpg

📄 frame007.25.00-07.30.00.txt

📄 frame007.30.00-07.35.00.txt

📄 frame007.35.00-07.40.00.txt

📄 frame007.40.00-07.45.00.txt

• Train set: 761

• Val set: 218

• Test set: 109

=> Total: 1088

❖ Step 4: Download dataset

Helmet Safety Detection dataset:

<https://drive.google.com/file/d/1twdtZEfcw4ghSZIiPDypJurZnNXzMO7R/view?usp=sharing>

1. Download with gdown:

```
1 !gdown '1twdtZEfcw4ghSZIiPDypJurZnNXzMO7R'  
2 !mkdir safety_helmet_dataset  
3 !unzip -q '/content/Safety_Helmet_Dataset.zip' -d '/content/safety_helmet_dataset'
```

```
/usr/local/lib/python3.7/dist-packages/gdown/cli.py:131: FutureWarning: Opti  
category=FutureWarning,  
Access denied with the following error:
```

```
Too many users have viewed or downloaded this file recently. Please  
try accessing the file again later. If the file you are trying to  
access is particularly large or is shared with many people, it may  
take up to 24 hours to be able to view or download the file. If you  
still can't access a file after 24 hours, contact your domain  
administrator.
```

You may still be able to access the file from the browser:

https://drive.google.com/uc?id=1bYytJdFa1D_r2km6ZZ8T99TnPlvyCAhf

```
unzip: cannot find or open helmet.zip, helmet.zip.zip or helmet.zip.ZIP.
```

2. Copy from google drive:

```
1 from google.colab import drive  
2  
3 drive.mount('/content/gdrive')
```

```
/content  
Mounted at /content/gdrive
```

```
1 !cp path_to_human_dataset.zip .
```

Method 1 might not work if the number of downloads is high.

❖ Step 4: Download dataset

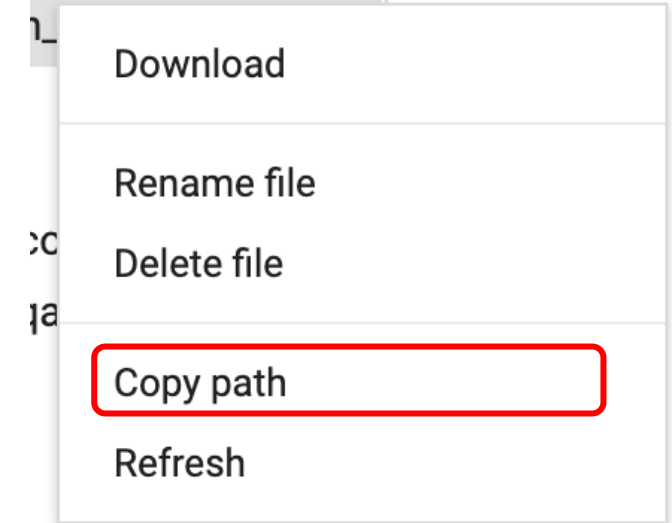
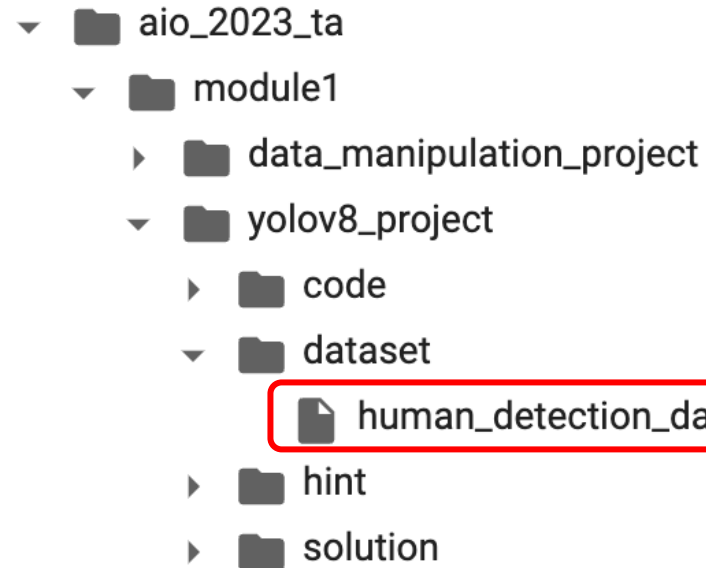
2. Copy from google drive example

```
1 from google.colab import drive
2
3 drive.mount('/content/gdrive')
```

/content

Mounted at /content/gdrive

```
1 !cp path_to_human_dataset.zip .
```



```
path_to_human_dataset.zip =  
/content/gdrive/MyDrive/Coordinate/aio_2023_ta/module1/yolov8_pro  
ject/dataset/human_detection_dataset.zip
```

YOLOv10

❖ Step 5: Prepare .yaml file

- data
 - hyps
 - images
 - scripts
 - Argoverse.yaml
 - GlobalWheat2020.yaml
 - Objects365.yaml
 - SKU-110K.yaml
 - VOC.yaml
 - VisDrone.yaml
 - coco.yaml**
 - coco128.yaml
 - helmet.yaml
 - market.yaml
 - xView.yaml

```
coco.yaml x
1 # YOLOv5 by Ultralytics, GPL-3.0 license
2 # COCO 2017 dataset http://cocodataset.org by Microsoft
3 # Example usage: python train.py --data coco.yaml
4 # parent
5 # |— yolov5
6 # |— datasets
7 # |   coco ← downloads here (20.1 GB)
8
9
10 # Train/val/test sets as 1) dir: path/to/imgs, 2) file: path/to/imgs.txt
11 path: ../datasets/coco # dataset root dir
12 train: train2017.txt # train images (relative to 'path') 118287 images
13 val: val2017.txt # val images (relative to 'path') 5000 images
14 test: test-dev2017.txt # 20288 of 40670 images, submit to https://compe
15
16 # Classes
17 nc: 80 # number of classes
18 names: ['person', 'bicycle', 'car', 'motorcycle', 'airplane', 'bus', 'tr
19         'fire hydrant', 'stop sign', 'parking meter', 'bench', 'bird', '
20         'elephant', 'bear', 'zebra', 'giraffe', 'backpack', 'umbrella',
21         'skis', 'snowboard', 'sports ball', 'kite', 'baseball bat', 'bas
22         'tennis racket', 'bottle', 'wine glass', 'cup', 'fork', 'knife',
23         'sandwich', 'orange', 'broccoli', 'carrot', 'hot dog', 'pizza',
24         'potted plant', 'bed', 'dining table', 'toilet', 'tv', 'laptop',
25         'microwave', 'oven', 'toaster', 'sink', 'refrigerator', 'book',
26         'hair drier', 'toothbrush'] # class names
27
```

Path to root folder

Path to train images
folder

Path to val images
folder

Path to test images
folder (optional)

Number of classes to
detect

Classes's name

❖ Step 5: Prepare .yaml file

▼ safety_helmet_dataset

▼ test

▶ images

▶ labels

▼ train

▶ images

▶ labels

▼ valid

▶ images

▶ labels

📄 README.dataset.txt

📄 README.roboflow.txt

📄 data.yaml

data.yaml X

```
1 train: ../train/images
2 val: ../valid/images
3 test: ../test/images
4
5 nc: 3
6 names: ['head', 'helmet', 'person']
7
8 roboflow:
9   workspace: dataperson
10  project: safety-helmet-dataset-uvh1t
11  version: 1
12  license: CC BY 4.0
13  url: https://universe.roboflow.com/dataperson/safety-helmet-dataset-uvh1t/dataset/1
```

Helmet Safety Dataset already provide us the .yaml file

❖ Step 5: Prepare .yaml file (Optional)

```
1 import yaml
2 import os
3
4 dataset_info = {
5     'train': './train/images',
6     'val': './val/images',
7     'path': os.path.abspath('./human_detection_dataset'),
8     'nc': 1,
9     'names': ['Human']
10 }
11
12 yaml_filepath = './human_detection_dataset/data.yaml'
13 with open(yaml_filepath, 'w') as f:
14     doc = yaml.dump(
15         dataset_info,
16         f,
17         default_flow_style=None,
18         sort_keys=False
19     )
```

**Dict of required
information**

❖ Step 6: Training

```
1 from ultralytics import YOLOv10
2
3 MODEL_PATH = 'yolov10n.pt'
4 model = YOLOv10(MODEL_PATH)
```

```
1 model.info()
```

YOLOv10n summary: 385 layers, 2775520 parameters, 2775504 gradients, 8.7 GFLOPs (385, 2775520, 2775504, 8.741504)

```
1 YAML_PATH = '../safety_helmet_dataset/data.yaml'
2 EPOCHS = 50
3 IMG_SIZE = 640
4 BATCH_SIZE = 256
5
6 model.train(data=YAML_PATH,
7             epochs=EPOCHS,
8             batch=BATCH_SIZE,
9             imgsz=IMG_SIZE)
```

	from	n	params	module
0	-1	1	464	ultralytics.nn.modules.conv.Conv
1	-1	1	4672	ultralytics.nn.modules.conv.Conv
2	-1	1	7360	ultralytics.nn.modules.block.C2f
3	-1	1	18560	ultralytics.nn.modules.conv.Conv
4	-1	2	49664	ultralytics.nn.modules.block.C2f
5	-1	1	9856	ultralytics.nn.modules.block.SCDown
6	-1	2	197632	ultralytics.nn.modules.block.C2f
7	-1	1	36096	ultralytics.nn.modules.block.SCDown
8	-1	1	460288	ultralytics.nn.modules.block.C2f
9	-1	1	164608	ultralytics.nn.modules.block.SPPF
10	-1	1	249728	ultralytics.nn.modules.block.PSA
11	-1	1	0	torch.nn.modules.upsampling.Upsample
12	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat
13	-1	1	148224	ultralytics.nn.modules.block.C2f
14	-1	1	0	torch.nn.modules.upsampling.Upsample
15	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat
16	-1	1	37248	ultralytics.nn.modules.block.C2f
17	-1	1	36992	ultralytics.nn.modules.conv.Conv
18	[-1, 13]	1	0	ultralytics.nn.modules.conv.Concat
19	-1	1	123648	ultralytics.nn.modules.block.C2f
20	-1	1	18048	ultralytics.nn.modules.block.SCDown
21	[-1, 10]	1	0	ultralytics.nn.modules.conv.Concat
22	-1	1	282624	ultralytics.nn.modules.block.C2fCIB
23	[16, 19, 22]	1	862498	ultralytics.nn.modules.head.v10Detect

YOLOv10n summary: 385 layers, 2708210 parameters, 2708194 gradients, 8.4 GFLOPs

❖ Step 6: Training

Log screen when training process is end.

Epoch	GPU_mem	box_om	cls_om	dfl_om	box_oo	cls_oo	dfl_oo	Instances	Size
48/50	3.22G	1.172	0.6252	1.274	1.219	0.6312	1.257	27	640: 100% ██████████ 48/48 [00:15<00:00, 3.01it/s]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100% ██████████	7/7 [00:03<00:00, 2.04it/s]
Epoch	GPU_mem	box_om	cls_om	dfl_om	box_oo	cls_oo	dfl_oo	Instances	Size
49/50	3.22G	1.138	0.6073	1.267	1.178	0.6224	1.251	22	640: 100% ██████████ 48/48 [00:17<00:00, 2.76it/s]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100% ██████████	7/7 [00:02<00:00, 2.63it/s]
Epoch	GPU_mem	box_om	cls_om	dfl_om	box_oo	cls_oo	dfl_oo	Instances	Size
50/50	3.24G	1.133	0.6103	1.262	1.169	0.6245	1.247	20	640: 100% ██████████ 48/48 [00:17<00:00, 2.69it/s]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100% ██████████	7/7 [00:02<00:00, 3.09it/s]

50 epochs completed in 0.318 hours.

Optimizer stripped from /content/yolov10/runs/detect/train/weights/last.pt, 5.8MB

Optimizer stripped from /content/yolov10/runs/detect/train/weights/best.pt, 5.8MB

Validating /content/yolov10/runs/detect/train/weights/best.pt...

Ultralytics YOLOv8.1.34 Python-3.10.12 torch-2.0.1+cu117 CUDA:0 (Tesla T4, 15102MiB)

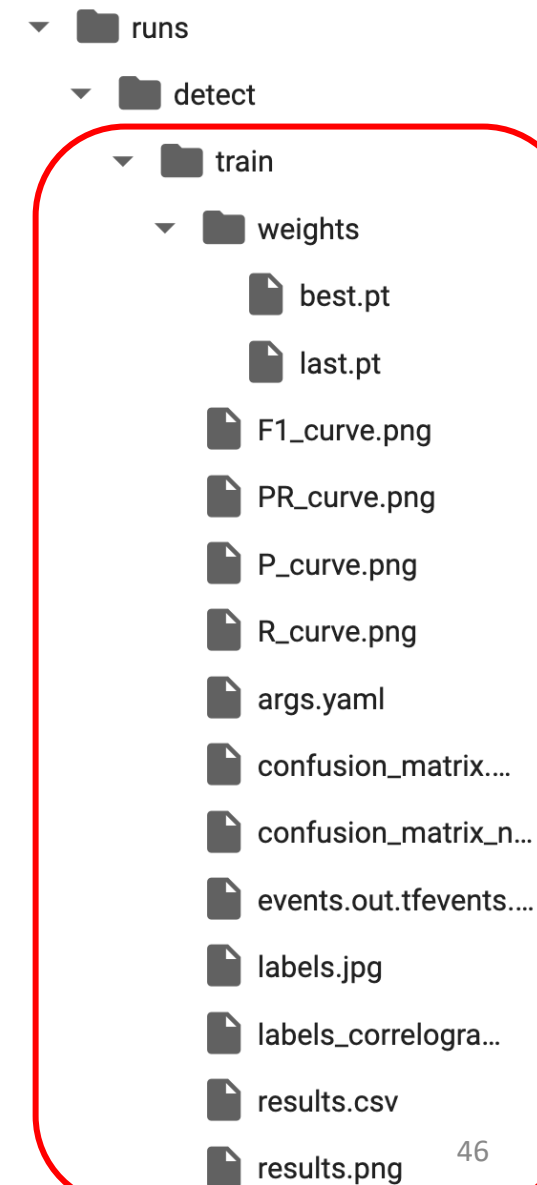
YOLOv10n summary (fused): 285 layers, 2695586 parameters, 0 gradients, 8.2 GFLOPs

Class	Images	Instances	Box(P	R	mAP50	mAP50-95):
all	218	586	0.826	0.766	0.851	0.443
head	218	27	0.806	0.741	0.802	0.395
helmet	218	285	0.874	0.852	0.924	0.467
person	218	274	0.797	0.704	0.828	0.465

Speed: 5.2ms preprocess, 4.9ms inference, 0.0ms loss, 0.2ms postprocess per image

Results saved to /content/yolov10/runs/detect/train

ultralytics.utils.metrics.DetMetrics object with attributes:



❖ Step 7: Predict

Epoch	GPU_mem	box_om	cls_om	dfl_om	box_oo	cls_oo	dfl_oo	Instances	Size
48/50	3.22G	1.172	0.6252	1.274	1.219	0.6312	1.257	27	640: 100% ██████████ 48/48 [00:15<00:00, 3.01it/s]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100% ██████████	7/7 [00:03<00:00, 2.04it/s] all
Epoch	GPU_mem	box_om	cls_om	dfl_om	box_oo	cls_oo	dfl_oo	Instances	Size
49/50	3.22G	1.138	0.6073	1.267	1.178	0.6224	1.251	22	640: 100% ██████████ 48/48 [00:17<00:00, 2.76it/s]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100% ██████████	7/7 [00:02<00:00, 2.63it/s] all
Epoch	GPU_mem	box_om	cls_om	dfl_om	box_oo	cls_oo	dfl_oo	Instances	Size
50/50	3.24G	1.133	0.6103	1.262	1.169	0.6245	1.247	20	640: 100% ██████████ 48/48 [00:17<00:00, 2.69it/s]
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95):	100% ██████████	7/7 [00:02<00:00, 3.09it/s] all

50 epochs completed in 0.318 hours.
Optimizer stripped from /content/yolov10/runs/detect/train/weights/last.pt, 5.8MB
Optimizer stripped from /content/yolov10/runs/detect/train/weights/best.pt, 5.8MB

Validating /content/yolov10/runs/detect/train/weights/best.pt...
Ultralytics YOLOv8.1.34 Python-3.10.12 torch-2.0.1+cu117 CUDA:0 (Tesla T4, 15102MiB)
YOLOv10n summary (fused): 285 layers, 2695586 parameters, 0 gradients, 8.2 GFLOPs

Class	Images	Instances	Box(P	R	mAP50	mAP50-95):
all	218	586	0.826	0.766	0.851	0.443
head	218	27	0.806	0.741	0.802	0.395
helmet	218	285	0.874	0.852	0.924	0.467
person	218	274	0.797	0.704	0.828	0.465

Speed: 5.2ms preprocess, 4.9ms inference, 0.0ms loss, 0.2ms postprocess per image
Results saved to /content/yolov10/runs/detect/train
ultralytics.utils.metrics.DetMetrics object with attributes:

Path to trained weight
file (.pt)

runs
└─ detect
 └─ predict
 └─ _108240740_beatles-abbeyroad
 └─ predict2
 └─ predict3
 └─ predict4
 └─ predict5

❖ Step 7: Predict

```
1 from google.colab.patches import cv2_imshow
2
3 TRAINED_MODEL_PATH = 'runs/detect/train/weights/best.pt'
4 model = YOLOv10(TRAINED_MODEL_PATH)
5
6 IMAGE_URL = 'IMG_URL'
7 CONF_THRESHOLD = 0.3
8 results = model.predict(source=IMAGE_URL,
9                           imgsiz=IMG_SIZE,
10                          conf=CONF_THRESHOLD)
11 annotated_img = results[0].plot()
12
13 cv2_imshow(annotated_img)
```

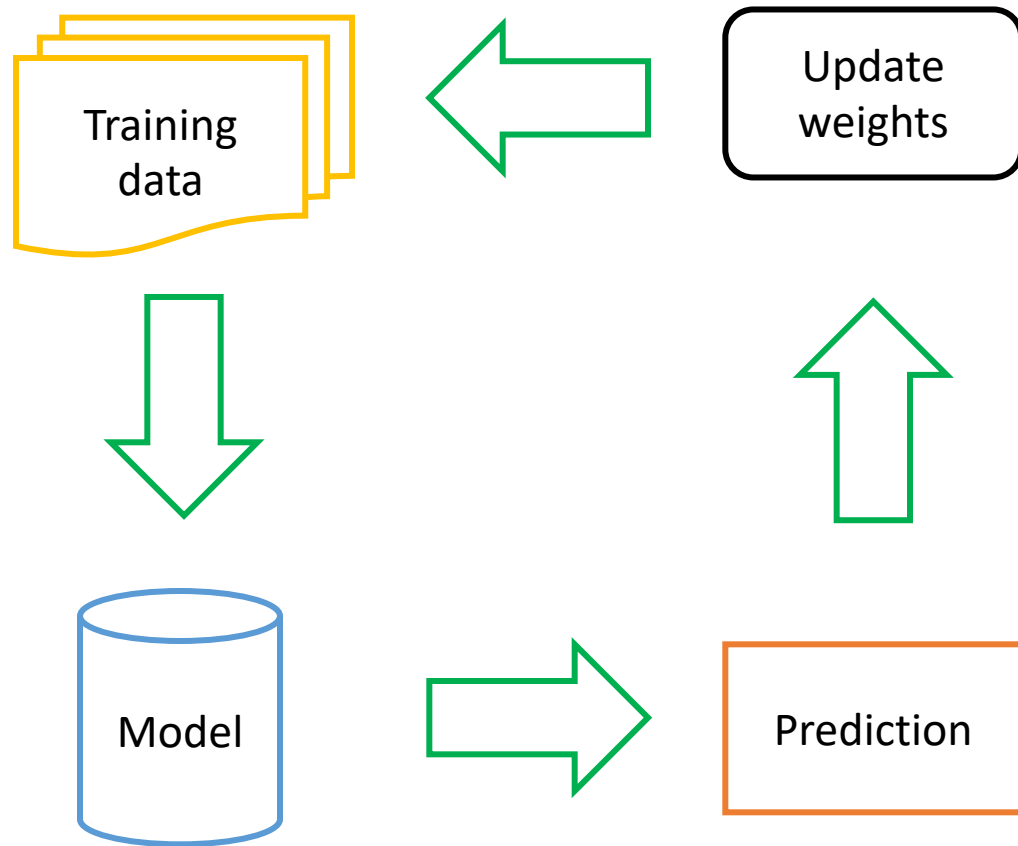


❖ Step 7: Predict

source	model(arg)	type	notes
image	'im.jpg'	str, Path	
URL	'https://ultralytics.com/images/bus.jpg'	str	
screenshot	'screen'	str	
PIL	Image.open('im.jpg')	PIL.Image	HWC, RGB
OpenCV	cv2.imread('im.jpg')	np.ndarray	HWC, BGR
numpy	np.zeros((640, 1280, 3))	np.ndarray	HWC
torch	torch.zeros(16, 3, 320, 640)	torch.Tensor	BCHW, RGB
CSV	'sources.csv'	str, Path	RTSP, RTMP, HTTP
video ✓	'vid.mp4'	str, Path	
directory ✓	'path/'	str, Path	
glob ✓	'path/*.jpg'	str	Use * operator
YouTube ✓	'https://youtu.be/Zgi9g1ksQHc'	str	
stream ✓	'rtsp://example.com/media.mp4'	str	RTSP, RTMP, HTTP

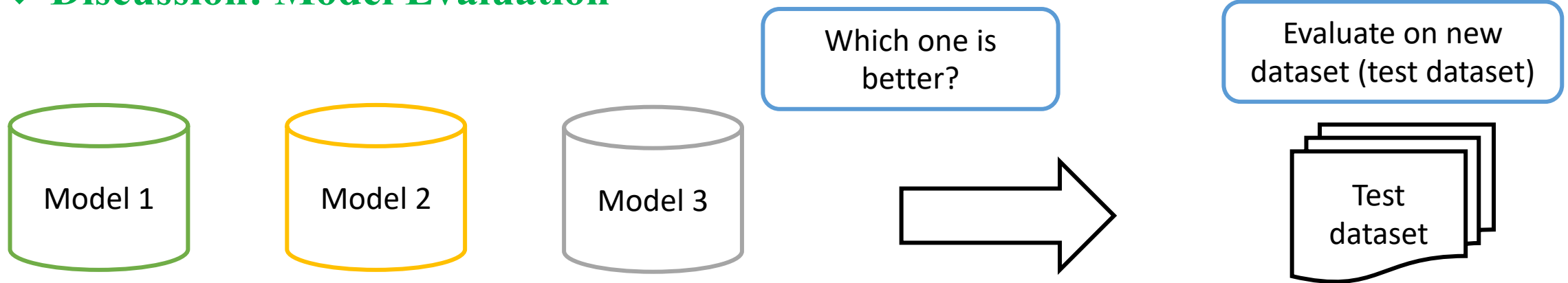
Other source options that we can input to run predict on YOLOv10.

❖ Discussion: Training command flags meaning



Flag	Meaning
img	The size of training image
batch	Number of data samples to read in 1 epoch
epochs	Number of iteration (number of times model "see" training data)
data	Path to .yaml file
weights	Path to pretrained model

❖ Discussion: Model Evaluation



```
1 TRAINED_MODEL_PATH = 'runs/detect/train/weights/best.pt'
2 model = YOLOv10(TRAINED_MODEL_PATH)
3
4 model.val(data=YAML_PATH,
5           imgs=IMG_SIZE,
6           split='test')
```

Ultralytics YOLOv8.1.34 🚀 Python-3.10.12 torch-2.0.1+cu117 CUDA:0 (Tesla T4, 15102MiB)

YOLOv10n summary (fused): 285 layers, 2695586 parameters, 0 gradients, 8.2 GFLOPs

val: Scanning /content/safety_helmet_dataset/test/labels... 109 images, 0 backgrounds, 0 corrupt: 100%|██████████| 109/109 [00:00<00:00, 901.31it/s] **val:**

/usr/lib/python3.10/multiprocessing/popen_fork.py:66: RuntimeWarning: os.fork() was called. os.fork() is incompatible with multithreaded code, and JAX is using forked multiprocessing. For more details, see https://jax.readthedocs.io/en/latest/faq.html#multiprocessing
self.pid = os.fork()

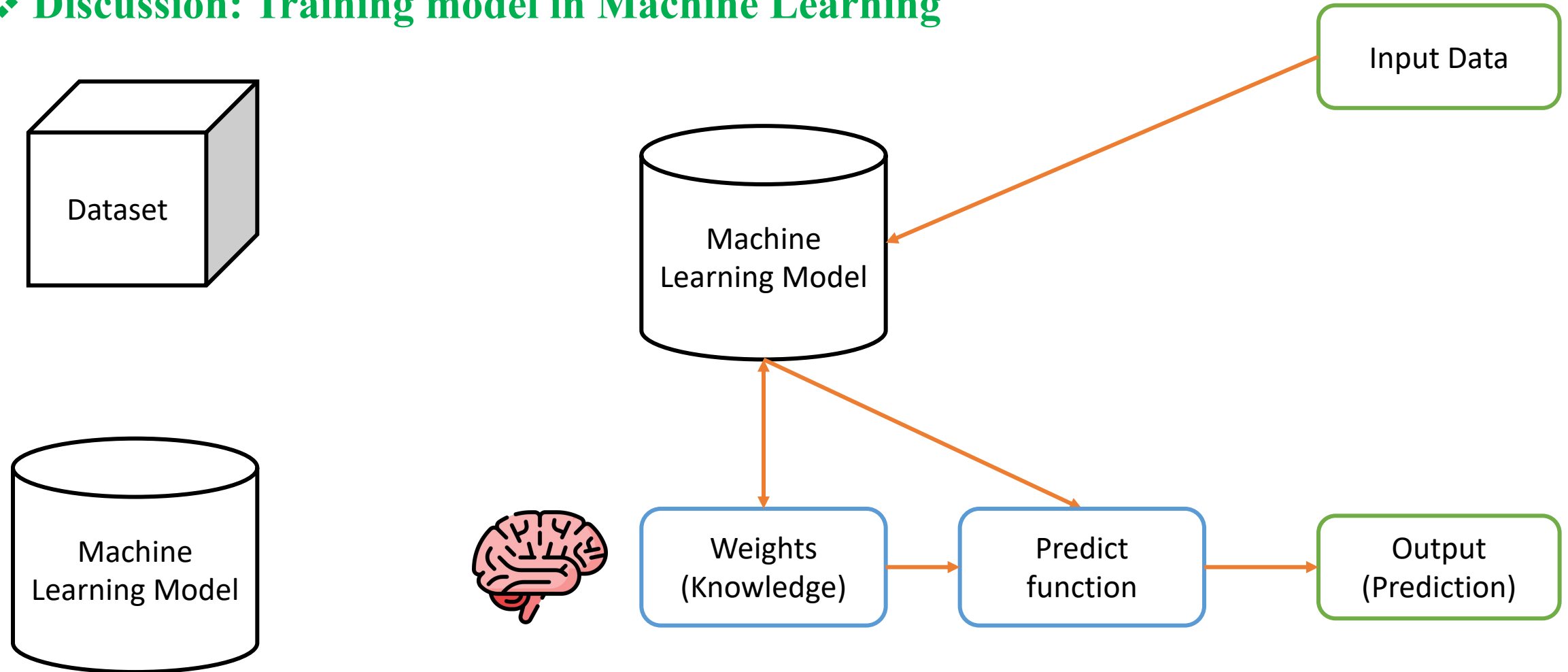
Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	7/7 [00:04<00:00, 1.41it/s]
all	109	320	0.768	0.738	0.816	0.417	
head	109	16	0.736	0.625	0.775	0.336	
helmet	109	162	0.818	0.864	0.893	0.465	
person	109	142	0.75	0.725	0.78	0.451	

Speed: 5.1ms preprocess, 12.7ms inference, 0.0ms loss, 1.0ms postprocess per image

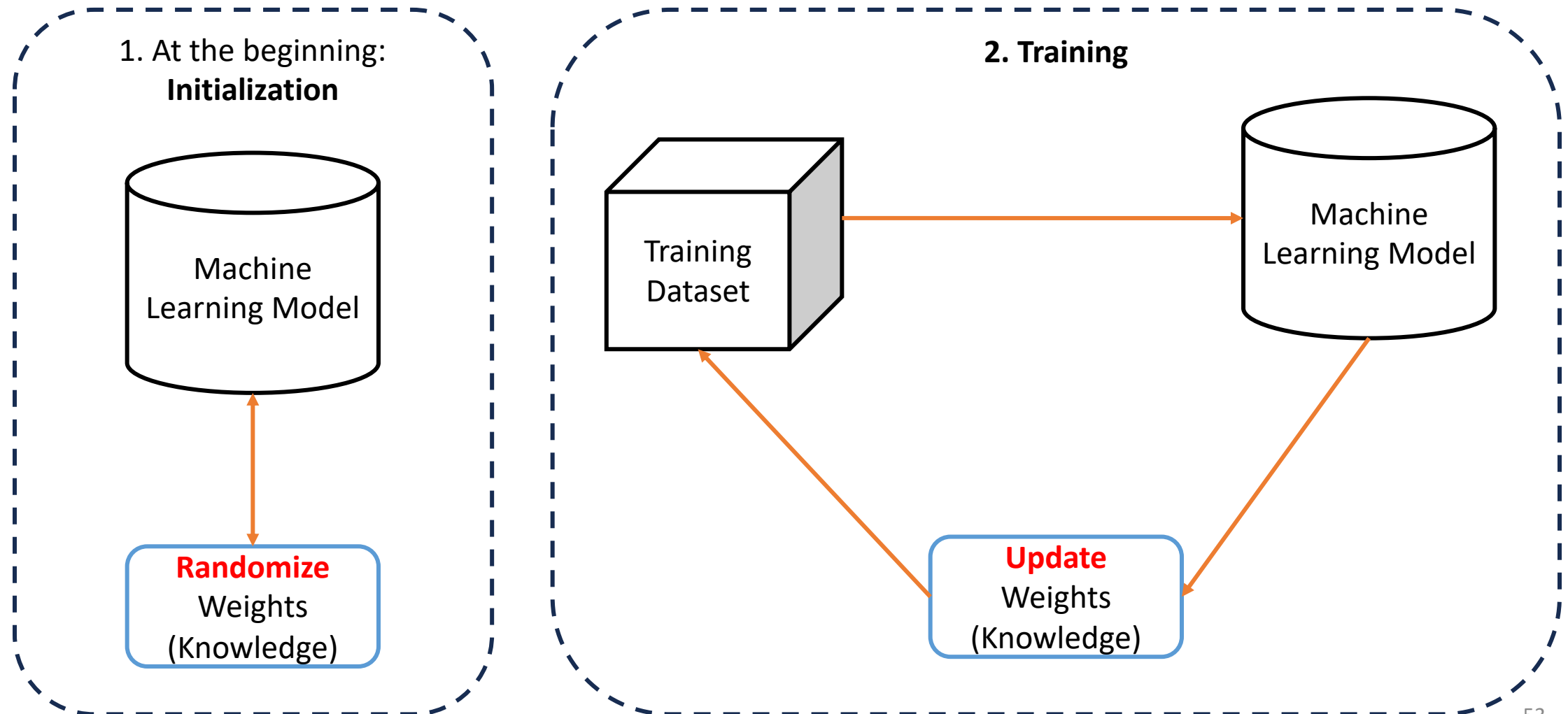
Results saved to /content/yolov10/runs/detect/val

ultralytics.utils.metrics.DetMetrics object with attributes:

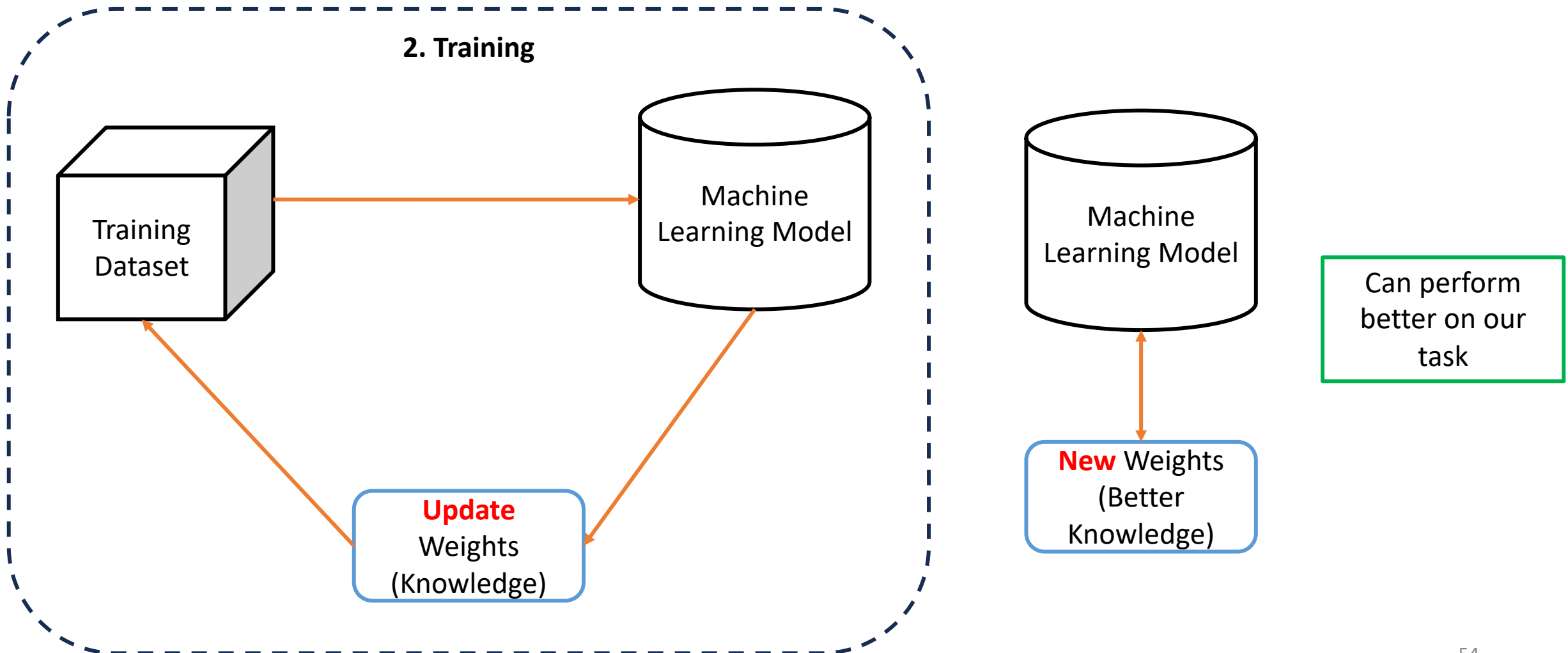
❖ Discussion: Training model in Machine Learning



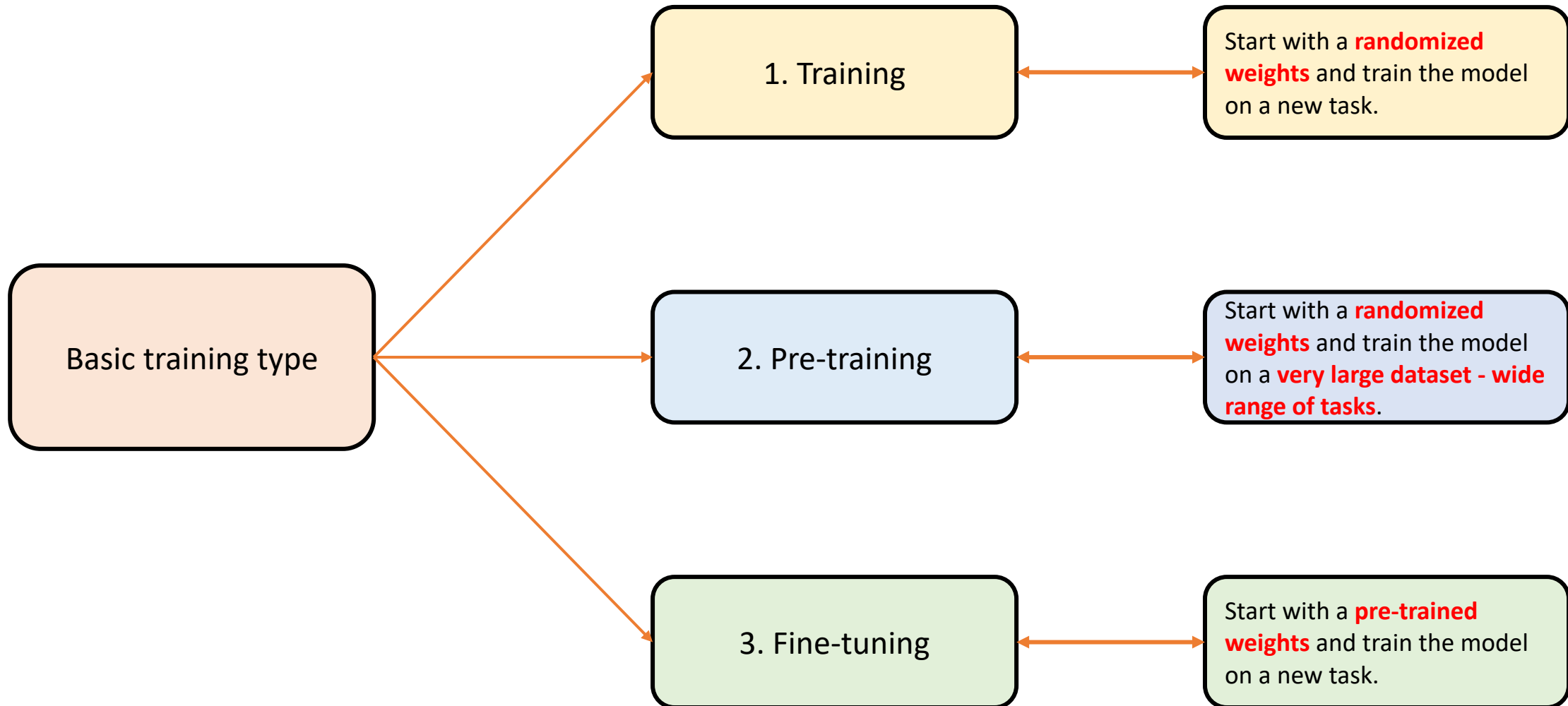
❖ Discussion: Training model in Machine Learning



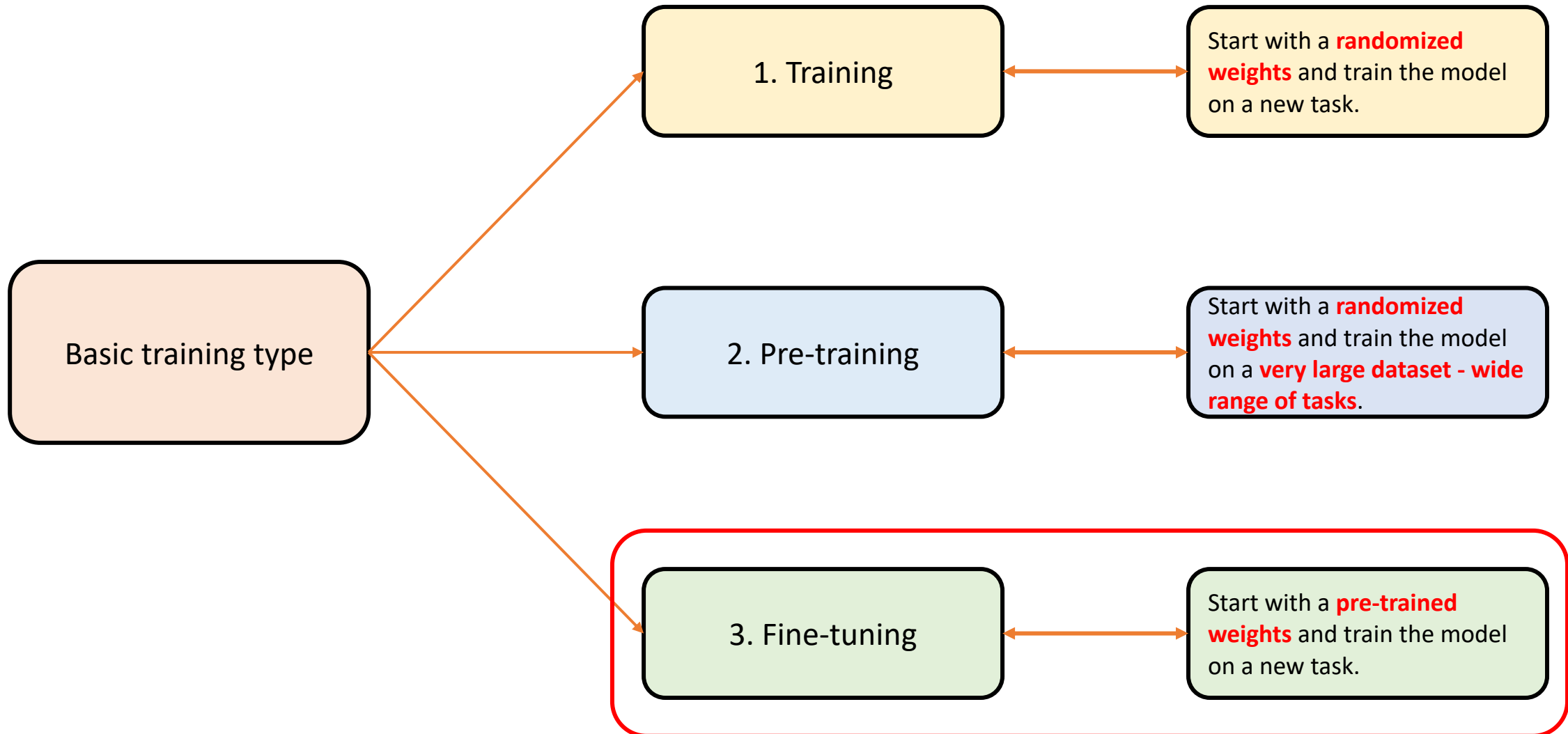
❖ Discussion: Training model in Machine Learning



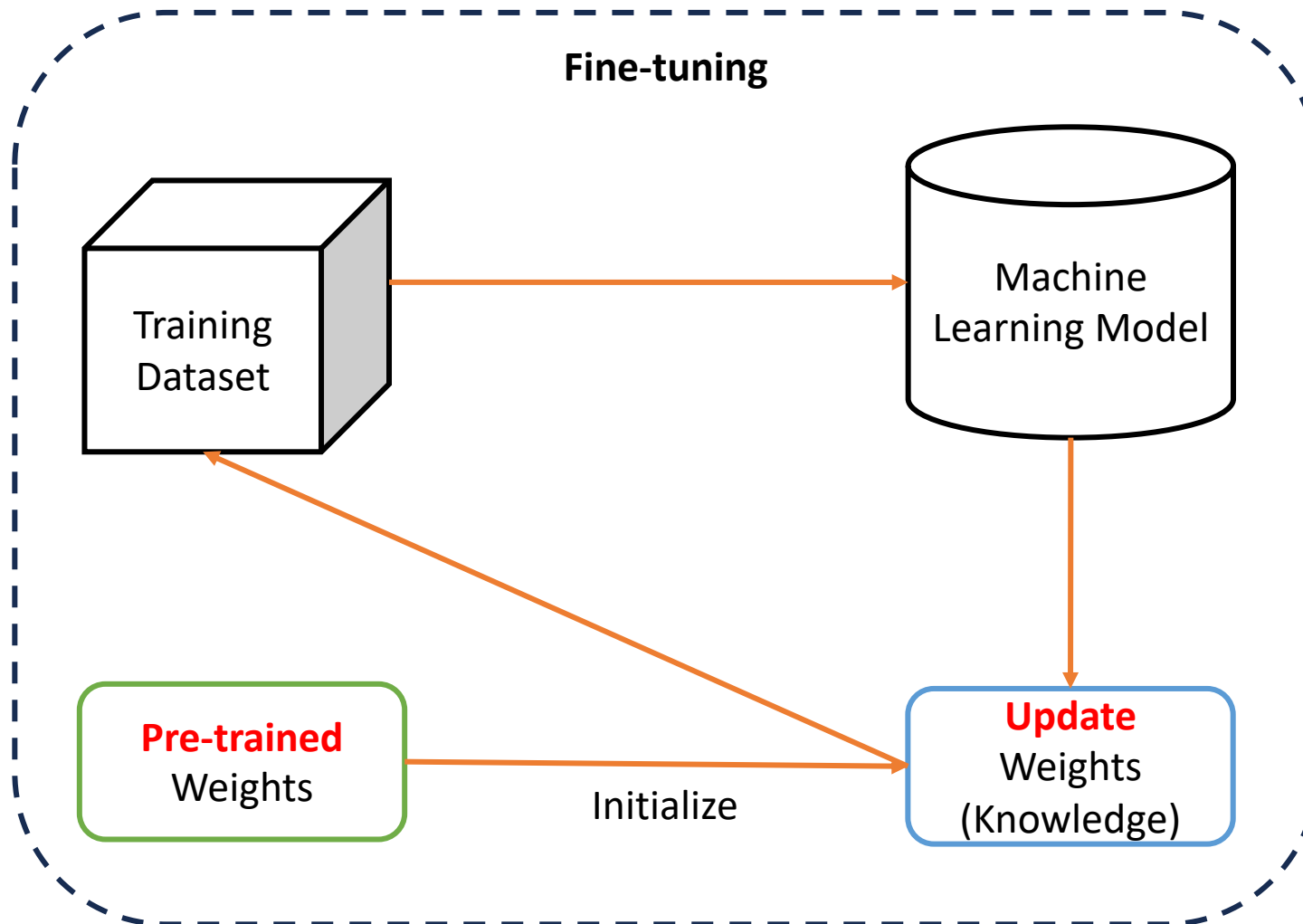
❖ Discussion: Training model in Machine Learning



❖ Discussion: Training model in Machine Learning



❖ Discussion: Fine-tuning



Linux Commands

Linux Commands (Colab)

❖ Introduction

2. Unzip dataset

```
1 !mkdir safety_helmet_dataset
```

```
1 !unzip -q '/content/Safety_Helmet_Dataset.zip' -d '/content/safety_helmet_dataset'
```

These commands are Linux commands

5. Download pre-trained weight

```
1 # Nano version: yolov10n.pt  
2 !wget https://github.com/THU-MIG/yolov10/releases/download/v1.1/yolov10n.pt
```

```
--2024-06-22 06:02:43-- https://github.com/THU-MIG/yolov10/releases/download/v1.1/yolov10n.pt  
Resolving github.com (github.com)... 140.82.112.4  
Connecting to github.com (github.com)|140.82.112.4|:443... connected.  
HTTP request sent, awaiting response... 302 Found
```

Linux Commands (Colab)

❖ Introduction



Microsoft Windows

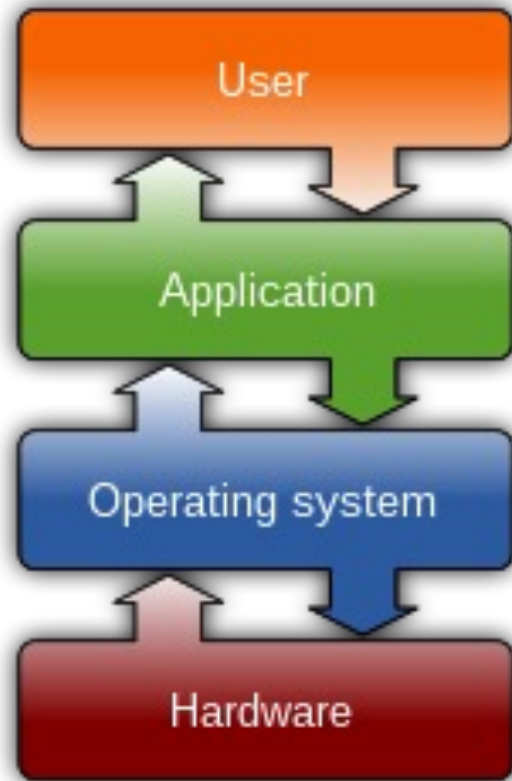


macOS

These are called Operating System (OS)

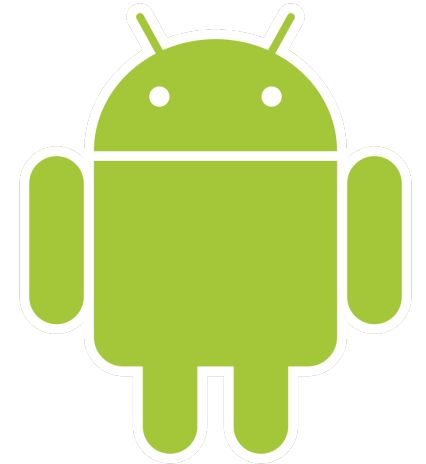
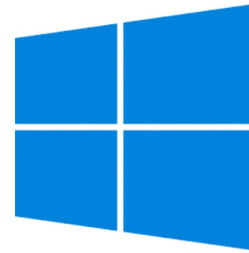
Linux Commands (Colab)

❖ Introduction



Operating System: A system software that manages computer hardware, software resources...

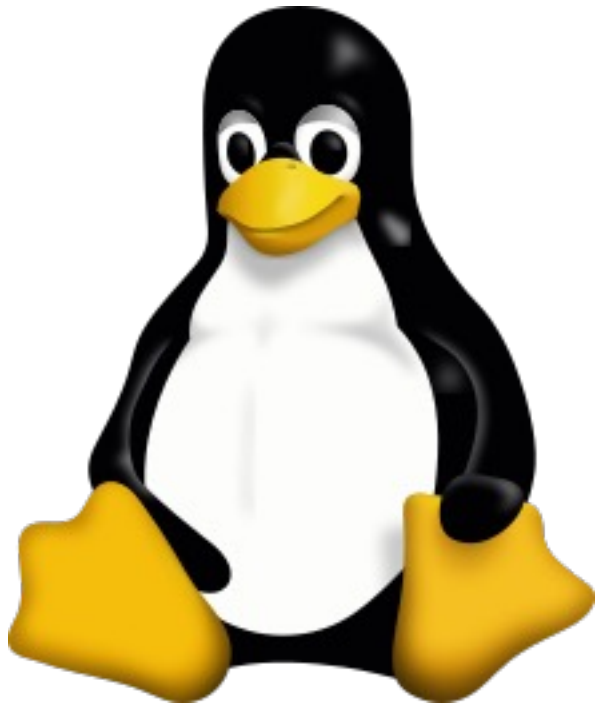
source: https://en.wikipedia.org/wiki/Operating_system



Some famous operating systems

Linux Commands (Colab)

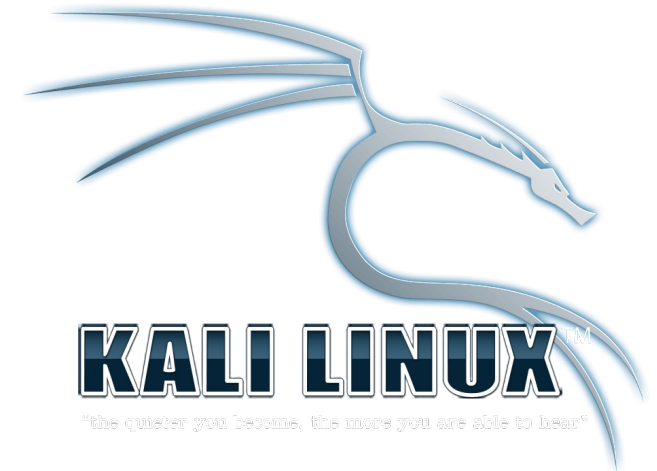
❖ Linux



Linux OS: A family of open-source Unix-like OSs based on the Linux Kernel



Linux Distributions: Operating System that based on Linux

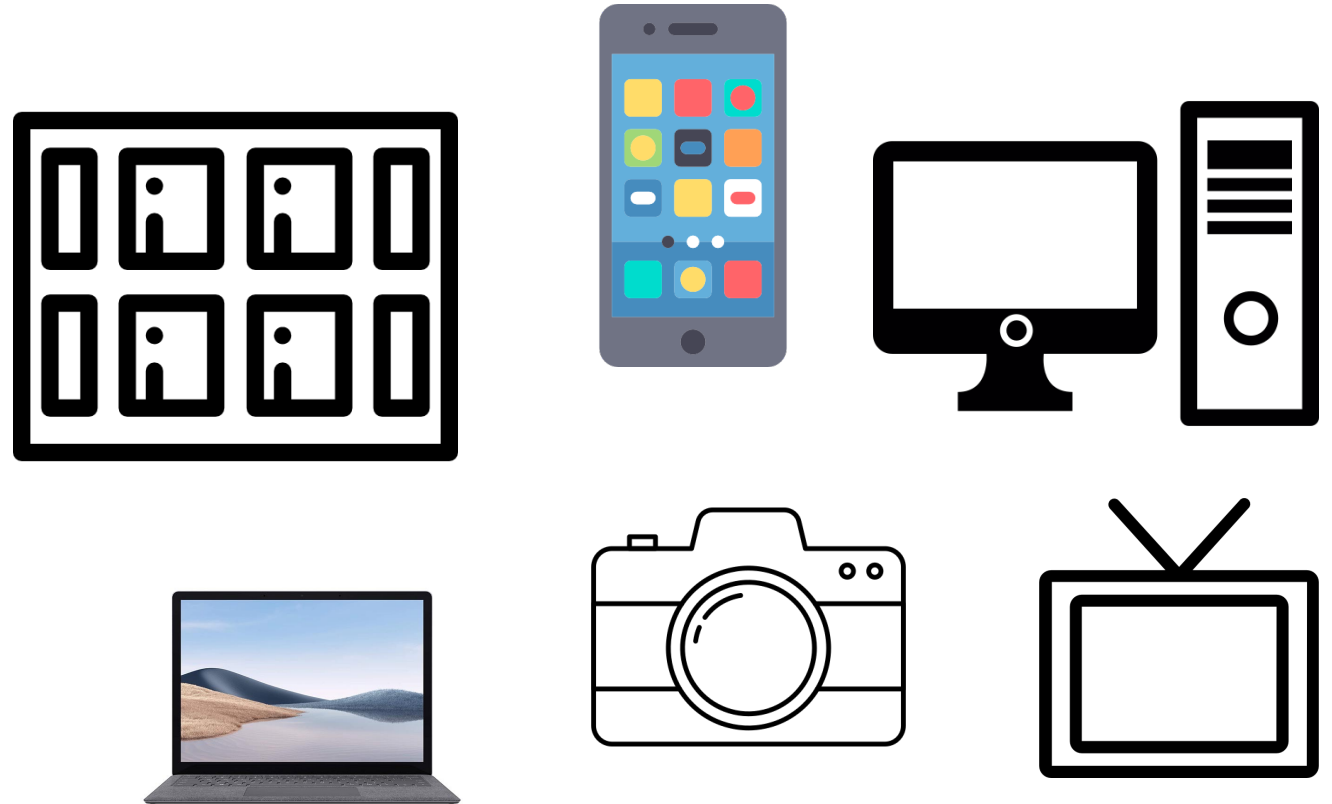


Linux Commands (Colab)

❖ Linux

Which systems use Linux?

- 100% of the top 500 supercomputers
- 36% of the top 10 million webserver
- 97% of the top 1 million webserver
- Nearly 100% of all Android devices
- 1.8% of desktops/laptops
- 15% of desktops/laptops that aren't running a Microsoft OS
- Roku STBs



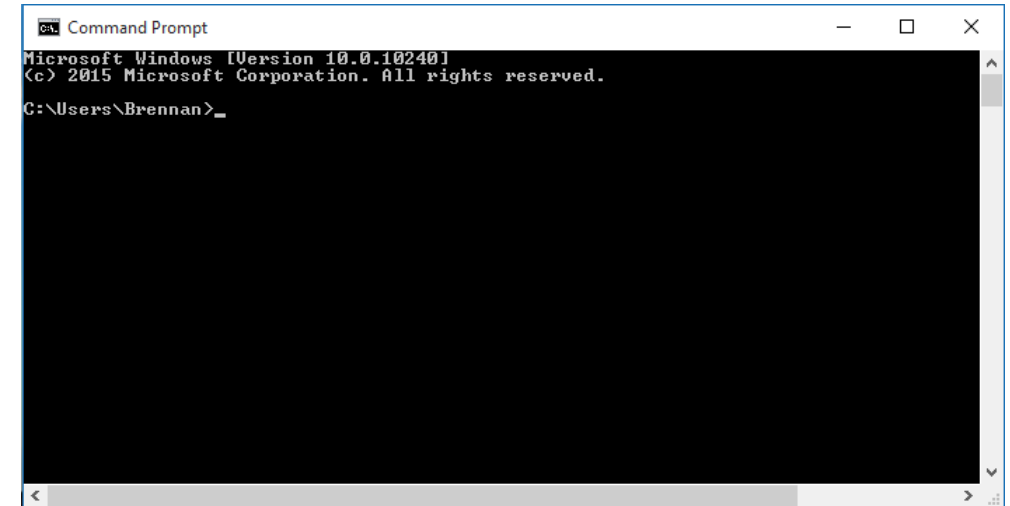
Lots of machines use Linux

Linux Commands (Colab)

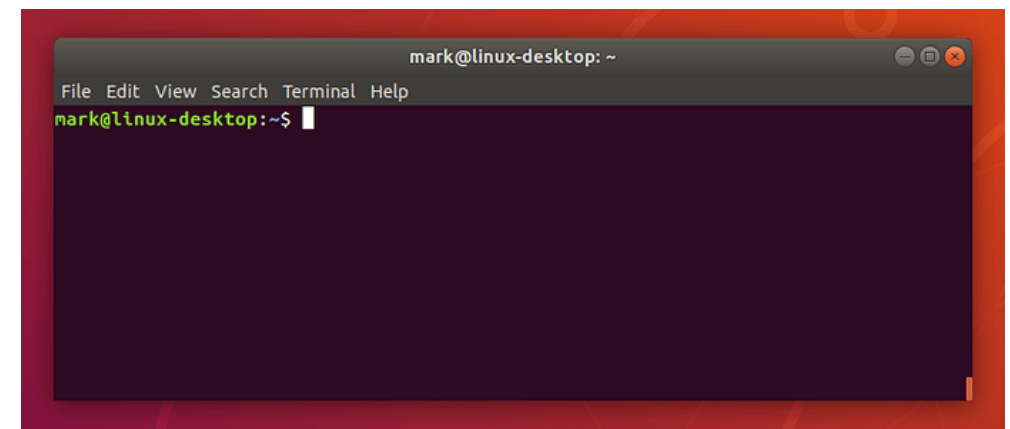
❖ Command Line Interface (CLI)

```
[root@localhost ~]# ping -q fa.wikipedia.org
PING text.pmtpa.wikimedia.org (208.80.152.2) 56(84) bytes of data.
^C
--- text.pmtpa.wikimedia.org ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 540.528/540.528/540.528/0.000 ms
[root@localhost ~]# pwd
/root
[root@localhost ~]# cd /var
[root@localhost var]# ls -la
total 72
drwxr-xr-x. 18 root root 4096 Jul 30 22:43 .
drwxr-xr-x. 23 root root 4096 Sep 14 20:42 ..
drwxr-xr-x.  2 root root 4096 May 14 00:15 account
drwxr-xr-x. 11 root root 4096 Jul 31 22:26 cache
drwxr-xr-x.  3 root root 4096 May 18 16:03 db
drwxr-xr-x.  3 root root 4096 May 18 16:03 empty
drwxr-xr-x.  2 root root 4096 May 18 16:03 games
drwxrwx--T.  2 root gdm  4096 Jun  2 18:39 gdm
drwxr-xr-x. 38 root root 4096 May 18 16:03 lib
drwxr-xr-x.  2 root root 4096 May 18 16:03 local
lrwxrwxrwx.  1 root root   11 May 14 00:12 lock -> ../run/lock
drwxr-xr-x. 14 root root 4096 Sep 14 20:42 log
lrwxrwxrwx.  1 root root   10 Jul 30 22:43 mail -> spool/mail
drwxr-xr-x.  2 root root 4096 May 18 16:03 nis
drwxr-xr-x.  2 root root 4096 May 18 16:03 opt
drwxr-xr-x.  2 root root 4096 May 18 16:03 preserve
drwxr-xr-x.  2 root root 4096 Jul  1 22:11 report
lrwxrwxrwx.  1 root root   6 May 14 00:12 run -> ../run
drwxr-xr-x. 14 root root 4096 May 18 16:03 spool
drwxrwxrwt.  4 root root 4096 Sep 12 23:50 tmp
drwxr-xr-x.  2 root root 4096 May 18 16:03 yp
[root@localhost var]# yum search wiki
Loaded plugins: langpacks, presto, refresh-packagekit, remove-with-leaves
rpmfusion-free-updates                                | 2.7 kB    00:00
rpmfusion-free-updates/primary_db                     | 206 kB    00:04
rpmfusion-nonfree-updates                             | 2.7 kB    00:00
updates/metalink                                       | 5.9 kB    00:00
updates                                                | 4.7 kB    00:00
updates/primary_db                                     73% [=====] 62 kB/s | 2.6 MB    00:15 ETA
```

Command-line interface (CLI): a text-based user interface used to run programs, interact with computer.



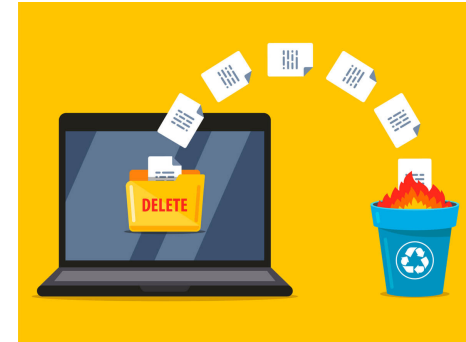
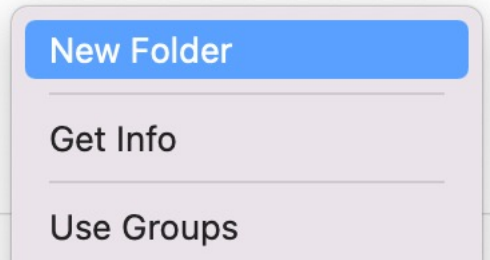
Windows Command Line (cmd)



Ubuntu Command Line (Terminal)

Linux Commands (Colab)

❖ Why CLI?



1. Create folder in traditional way

```
thangduong@Duongs-MacBook-Pro Desktop % mkdir new_folder  
thangduong@Duongs-MacBook-Pro Desktop %
```



new_folder

2. Create folder using command line



Can do many process conveniently

Linux Commands (Colab)

❖ Linux Command Line

```
[thangduong@Duongs-MacBook-Pro Desktop % mkdir -h  
mkdir: illegal option -- h  
usage: mkdir [-pv] [-m mode] directory_name ...  
thangduong@Duongs-MacBook-Pro Desktop %
```

```
usage: mkdir [-pv] [-m mode] directory_name ...  
[thangduong@Duongs-MacBook-Pro Desktop % mkdir new_folder  
thangduong@Duongs-MacBook-Pro Desktop %
```

argument: Input given to a command line

flag: An option for the command,
can be started with (-) or (--)

Linux Commands (Colab)

❖ Common Linux Commands: ls

```
[(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff          examples
CONTRIBUTING.md      mkdocs.yml
LICENSE               requirements.txt
MANIFEST.in           setup.cfg
README.md              setup.py
README.zh-CN.md       tests
docker                ultralytics
docs
```

❖ **Command:** `ls`

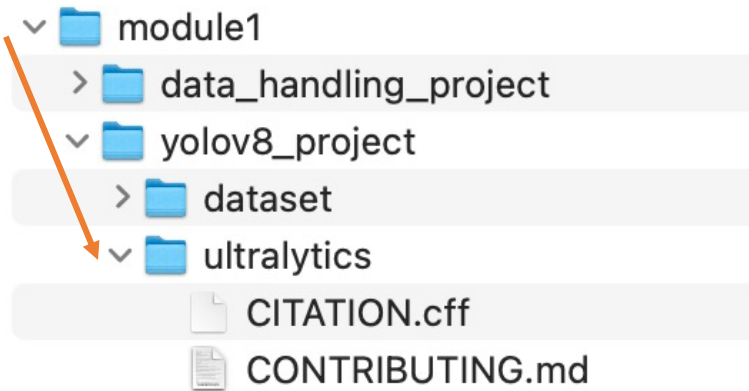
❖ **Usage:** To list all file/folder names in current directory.

< > ultralytics	
Name	
	CITATION.cff
	CONTRIBUTING.md
> 	docker
> 	docs
> 	examples
	LICENSE
	MANIFEST.in
	mkdocs.yml
	README.md
	README.zh-CN.md
	requirements.txt
	setup.cfg
	setup.py
> 	tests
> 	ultralytics

Linux Commands (Colab)

❖ Common Linux Commands: cd

```
(base) thangduong@Duongs-MacBook-Pro module1 % ls
data_handling_project  yolov8_project
(base) thangduong@Duongs-MacBook-Pro module1 % cd yolov8_project
(base) thangduong@Duongs-MacBook-Pro yolov8_project % ls
dataset                ultralytics
(base) thangduong@Duongs-MacBook-Pro yolov8_project % cd ultralytics
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff          README.zh-CN.md      requirements.txt
CONTRIBUTING.md      docker               setup.cfg
LICENSE               docs                 setup.py
MANIFEST.in           examples             tests
README.md             mkdocs.yml           ultralytics
(base) thangduong@Duongs-MacBook-Pro ultralytics %
```



E.g: Move from
module1 to ultralytics

❖ **Command:** `cd`

❖ **Usage:** “change directory”, move to the specified directory.

```
thangduong@Duongs-MacBook-Pro data % ls
Argoverse.yaml      VOC.yaml             hyps
GlobalWheat2020.yaml VisDrone.yaml        images
Objects365.yaml     coco.yaml            scripts
SKU-110K.yaml       coco128.yaml         xView.yaml
thangduong@Duongs-MacBook-Pro data % cd ..
thangduong@Duongs-MacBook-Pro yolov5 % ls
CONTRIBUTING.md    export.py            train.py
LICENSE             hubconf.py          tutorial.ipynb
README.md           models              utils
data                requirements.txt     val.py
detect.py           setup.cfg
thangduong@Duongs-MacBook-Pro yolov5 %
```

Move to “1 level up” directory

```
thangduong@Duongs-MacBook-Pro yolov5 % cd ..
thangduong@Duongs-MacBook-Pro ~ % pwd
/Users/thangduong
thangduong@Duongs-MacBook-Pro ~ %
```

Move to home directory

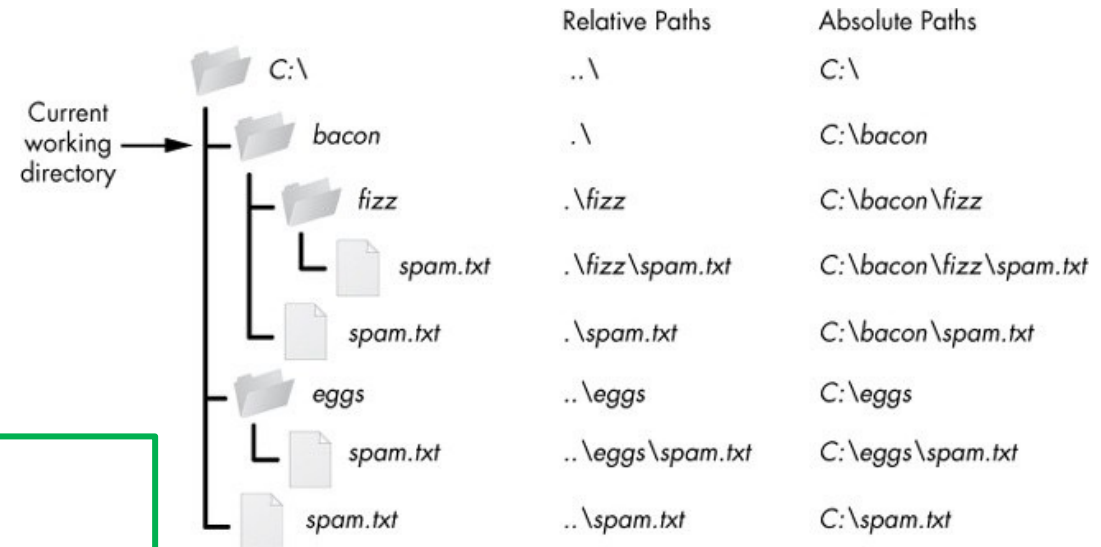
Linux Commands (Colab)

❖ Common Linux Commands: pwd

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % pwd
/Users/thangduong/Desktop/aio2023/module1/yolov8_project/ultralytics
(base) thangduong@Duongs-MacBook-Pro ultralytics % cd ..
(base) thangduong@Duongs-MacBook-Pro yolov8_project % pwd
/Users/thangduong/Desktop/aio2023/module1/yolov8_project
(base) thangduong@Duongs-MacBook-Pro yolov8_project %
```

❖ **Command:** `pwd`

❖ **Usage:** “Print Working Directory”, output the absolute path of current directory.



Relative Path vs Absolute Path

Linux Commands (Colab)

❖ Common Linux Commands: cp and rm

❖ **Command:** cp

❖ **Usage:** copy file/folder to directory.

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls examples
README.md          hub.ipynb
YOLOv8-CPP-Inference tutorial.ipynb
YOLOv8-OpenCV-ONNX-Python
(base) thangduong@Duongs-MacBook-Pro ultralytics % cp ./examples/tutorial.ipynb .
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff      docker      setup.py
CONTRIBUTING.md  docs        tests
LICENSE           examples    tutorial.ipynb
MANIFEST.in       mkdocs.yml  ultralytics
README.md         requirements.txt
README.zh-CN.md   setup.cfg
(base) thangduong@Duongs-MacBook-Pro ultralytics %
```

E.g: Copy tutorial.ipynb from ./examples to current directory.

❖ **Command:** rm

❖ **Usage:** delete file/folder of specified directory.

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff      docker      setup.py
CONTRIBUTING.md  docs        tests
LICENSE           examples    tutorial.ipynb
MANIFEST.in       mkdocs.yml  ultralytics
README.md         requirements.txt
README.zh-CN.md   setup.cfg
(base) thangduong@Duongs-MacBook-Pro ultralytics % rm tutorial.ipynb
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff      README.zh-CN.md  requirements.txt
CONTRIBUTING.md  docker           setup.cfg
LICENSE          docs            setup.py
MANIFEST.in      examples        tests
README.md        mkdocs.yml      ultralytics
(base) thangduong@Duongs-MacBook-Pro ultralytics %
```

E.g: Delete tutorial.ipynb in current directory.

Linux Commands (Colab)

❖ Common Linux Commands: mv and mkdir

❖ **Command:** mv

❖ **Usage:** Move a file/directory to the other directory.

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls ./docs
CNAME                hub                robots.txt
README.md            index.md           stylesheets
SECURITY.md          models            tasks
assets              modes              usage
build_reference.py   overrides          yolov5
datasets            quickstart.md
help                reference
```

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % mv ./docs/yolov5 .
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff         docker             setup.py
CONTRIBUTING.md     docs              tests
LICENSE              examples           ultralytics
MANIFEST.in          mkdocs.yml        yolov5
README.md            requirements.txt
README.zh-CN.md      setup.cfg
```

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls ./docs
CNAME                help              quickstart.md
README.md            hub              reference
SECURITY.md          index.md         robots.txt
assets              models           stylesheets
build_reference.py   modes           tasks
datasets            overrides        usage
(base) thangduong@Duongs-MacBook-Pro ultralytics %
```

❖ **Command:** mkdir

❖ **Usage:** Create a folder.

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff         README.zh-CN.md   requirements.txt
CONTRIBUTING.md     docker            setup.cfg
LICENSE              docs              setup.py
MANIFEST.in          examples          tests
README.md            mkdocs.yml        ultralytics
```

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % mkdir abc
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff         abc               setup.cfg
CONTRIBUTING.md     docker           setup.py
LICENSE              docs            tests
MANIFEST.in          examples        ultralytics
README.md            mkdocs.yml
README.zh-CN.md      requirements.txt
(base) thangduong@Duongs-MacBook-Pro ultralytics %
```

Linux Commands (Colab)

❖ Common Linux Commands: touch and cat

- ❖ **Command:** touch
- ❖ **Usage:** Create a file.

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff      README.zh-CN.md  requirements.txt
CONTRIBUTING.md  docker           setup.cfg
LICENSE           docs            setup.py
MANIFEST.in       examples        tests
README.md         mkdocs.yml      ultralytics
(base) thangduong@Duongs-MacBook-Pro ultralytics % touch abc.txt
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff      abc.txt          setup.cfg
CONTRIBUTING.md  docker          setup.py
LICENSE           docs            tests
MANIFEST.in       examples        ultralytics
README.md         mkdocs.yml
README.zh-CN.md   requirements.txt
(base) thangduong@Duongs-MacBook-Pro ultralytics %
```

- ❖ **Command:** cat
- ❖ **Usage:** Display, create file.

```
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff      README.zh-CN.md  requirements.txt
CONTRIBUTING.md  docker           setup.cfg
LICENSE           docs            setup.py
MANIFEST.in       examples        tests
README.md         mkdocs.yml      ultralytics
(base) thangduong@Duongs-MacBook-Pro ultralytics % cat > abc.txt
Hello World
^C
(base) thangduong@Duongs-MacBook-Pro ultralytics % ls
CITATION.cff      abc.txt          setup.cfg
CONTRIBUTING.md  docker          setup.py
LICENSE           docs            tests
MANIFEST.in       examples        ultralytics
README.md         mkdocs.yml
README.zh-CN.md   requirements.txt
(base) thangduong@Duongs-MacBook-Pro ultralytics % cat abc.txt
Hello World
(base) thangduong@Duongs-MacBook-Pro ultralytics %
```


QUIZ

Labelling

❖ Introduction

YOLOv10 Dataset folder format

- ▼ safety_helmet_dataset
 - ▼ test
 - ▶ images
 - ▶ labels
 - ▼ train
 - ▶ images
 - ▶ labels
 - ▼ valid
 - ▶ images
 - ▶ labels

- ▶ frame007.25.00-07.30.00.jpg
- ▶ frame007.30.00-07.35.00.jpg
- ▶ frame007.35.00-07.40.00.jpg
- ▶ frame007.40.00-07.45.00.jpg

- ▶ frame007.25.00-07.30.00.txt
- ▶ frame007.30.00-07.35.00.txt
- ▶ frame007.35.00-07.40.00.txt
- ▶ frame007.40.00-07.45.00.txt

- What if we don't have "labels"?
- What if we want to create a new dataset?
- ...



ABC
family friendly workplace ...



Vecteezy
Construction Stock Photos, Images and ...



Safety Culture
What is Commercial Construction ...

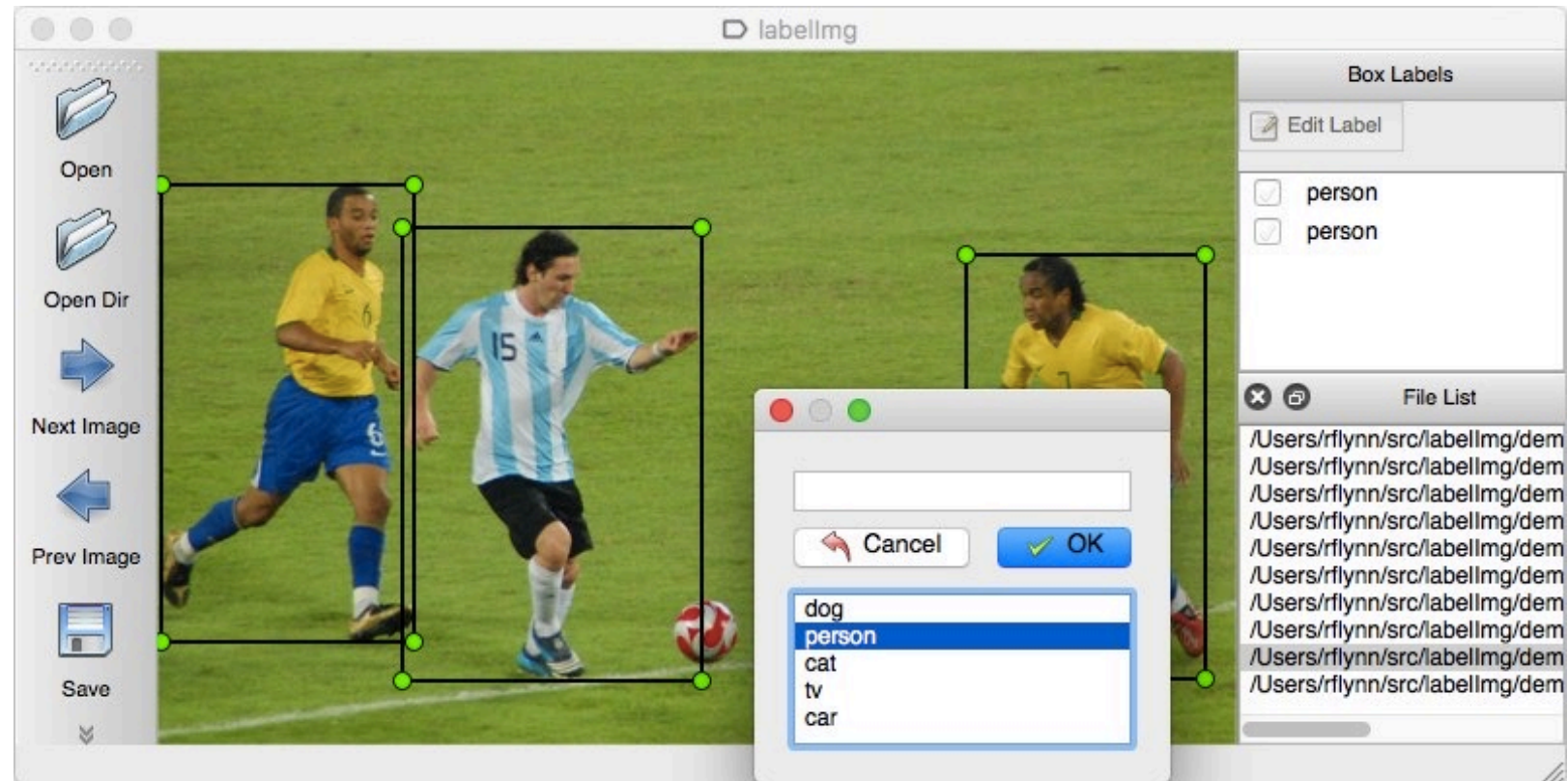


ILO
Construction sector | International ...

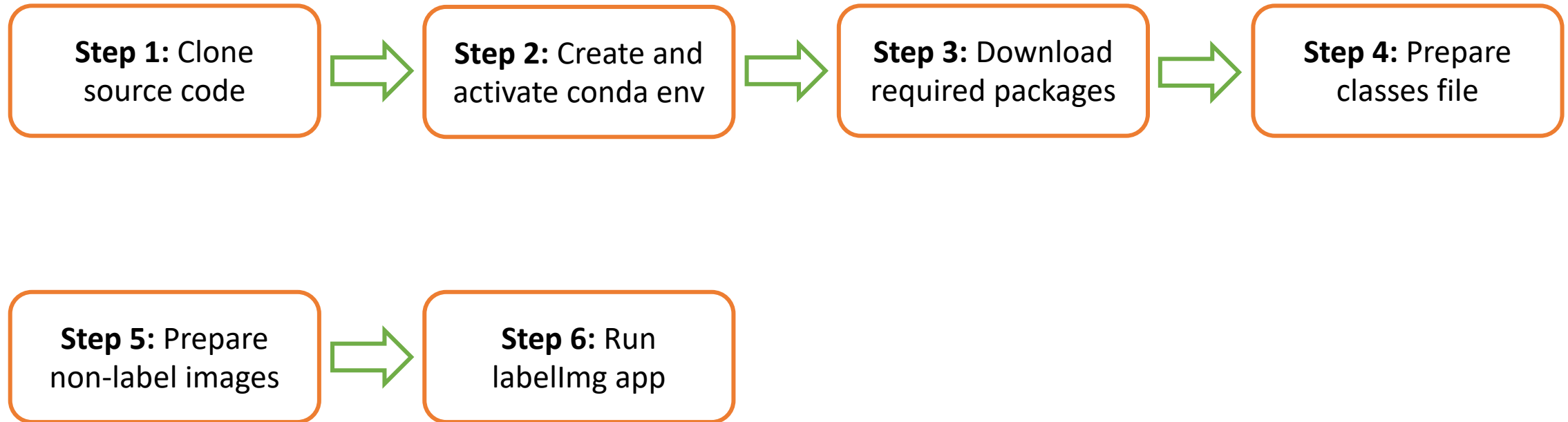
❖ Introduction



A graphical image
annotation tool



❖ labellmg Installation and Usage



❖ labelImg Installation

Installation

Get from PyPI but only python3.0 or above

This is the simplest (one-command) install method on modern Linux distributions such as Ubuntu and Fedora.

```
pip3 install labelImg
labelImg
labelImg [IMAGE_PATH] [PRE-DEFINED CLASS FILE]
```

Build from source

Linux/Ubuntu/Mac requires at least [Python 2.6](#) and has been tested with [PyQt 4.8](#). However, [Python 3 or above](#) and [PyQt5](#) are strongly recommended.

Ubuntu Linux

Python 3 + Qt5

```
sudo apt-get install pyqt5-dev-tools
sudo pip3 install -r requirements/requirements-linux-python3.txt
make qt5py3
python3 labelImg.py
python3 labelImg.py [IMAGE_PATH] [PRE-DEFINED CLASS FILE]
```

❖ labelImg Installation

Step 1: Clone source code

The screenshot shows the GitHub repository for labelImg. The repository is owned by 'Isell' and is a 'Public archive'. It has 401 watchers, 6.2k forks, and 22.1k stars. The repository is currently on the 'master' branch, with 5 branches and 25 tags. The file list includes: .github, build-tools, data, demo, libs, readme, requirements, resources, tests, tools, .gitignore, and CONTRIBUTING.rst. The 'About' section states that labelImg is now part of the Label Studio community and is no longer actively being developed. It provides a link to a YouTube video and a list of tags: recognition, tools, deep-learning, detection, annotations, python3, imagenet, image-classification, and python2. The repository also has a README, MIT license, and activity feed.

The screenshot shows the 'Clone' dropdown menu for the labelImg repository. It has two tabs: 'Local' and 'Codespaces'. The 'Clone' button is highlighted. Below it, there are three options: 'HTTPS', 'SSH', and 'GitHub CLI'. The 'HTTPS' option is selected, and the URL 'https://github.com/HumanSignal/labelImg.git' is displayed. There is a 'Clone using the web URL.' button. Below that, there are two more options: 'Open with GitHub Desktop' and 'Download ZIP'.

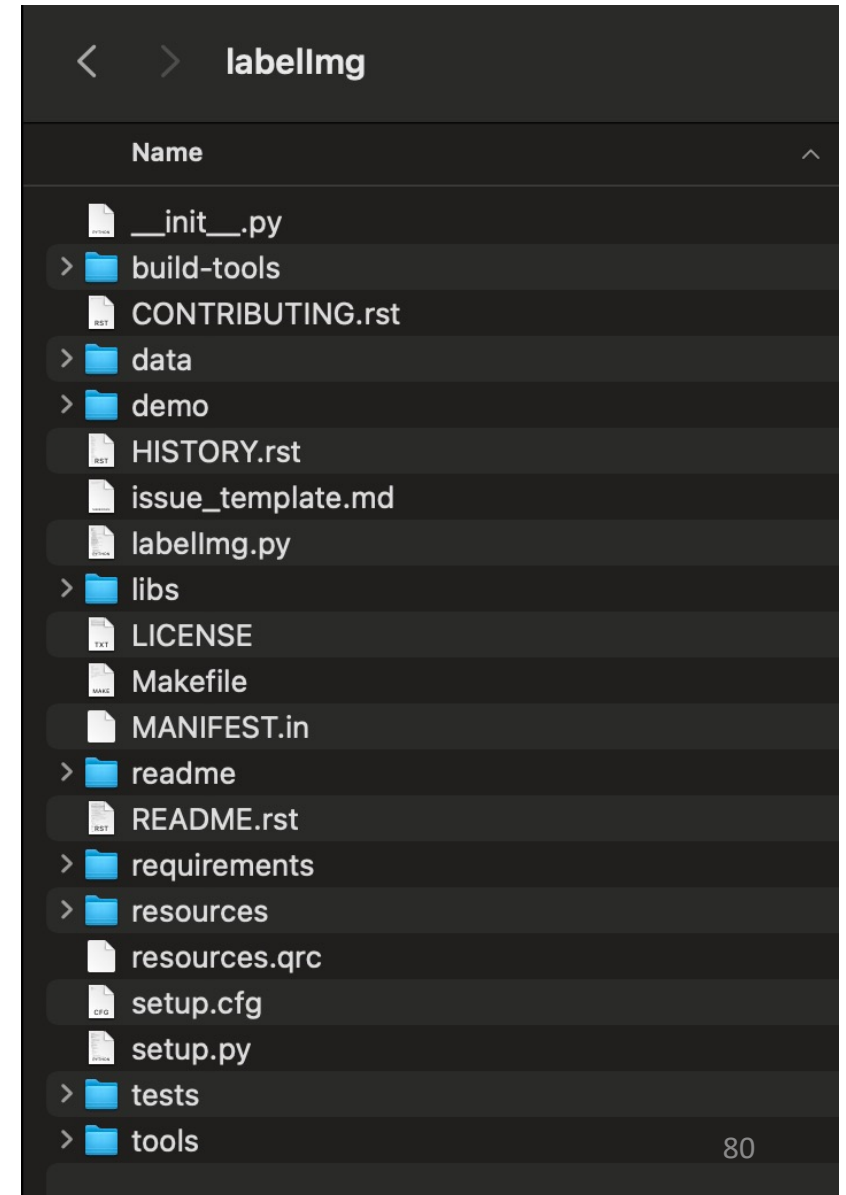
Step 2: Copy link and do git clone

❖ labellmg Installation

```
[(base) thangduong@Duongs-MacBook-Pro Project_YOLOv10 % git clone https://github.com/HumanSignal/labellmg.git
Cloning into 'labellmg'...
remote: Enumerating objects: 2097, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 2097 (delta 0), reused 4 (delta 0), pack-reused 2090
Receiving objects: 100% (2097/2097), 237.14 MiB | 7.39 MiB/s, done.
Resolving deltas: 100% (1245/1245), done.
(base) thangduong@Duongs-MacBook-Pro Project_YOLOv10 %
```

Use git clone command to
get labellmg src

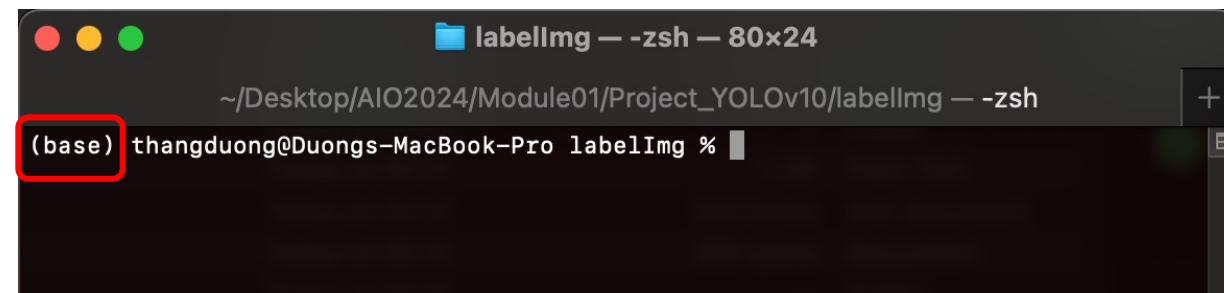
```
[(base) thangduong@Duongs-MacBook-Pro Project_YOLOv10 % cd labellmg
[(base) thangduong@Duongs-MacBook-Pro labellmg % ls
CONTRIBUTING.rst      build-tools      requirements
HISTORY.rst           data             resources
LICENSE               demo            resources.qrc
MANIFEST.in           issue_template.md setup.cfg
Makefile              labellmg.py     setup.py
README.rst            libs            tests
__init__.py           readme          tools
(base) thangduong@Duongs-MacBook-Pro labellmg %
```



labellmg

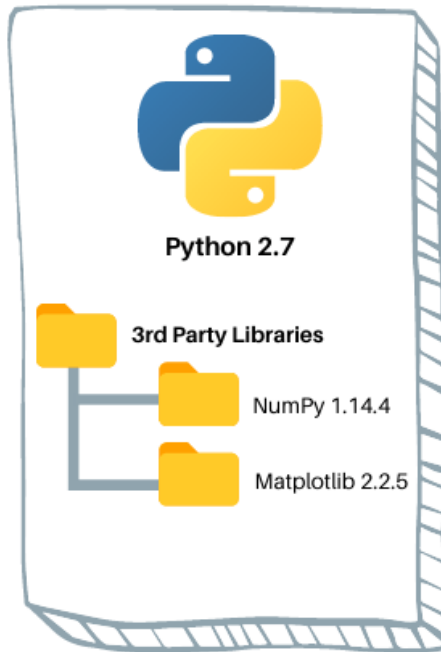
❖ labellmg Installation

Step 2: Download Anaconda

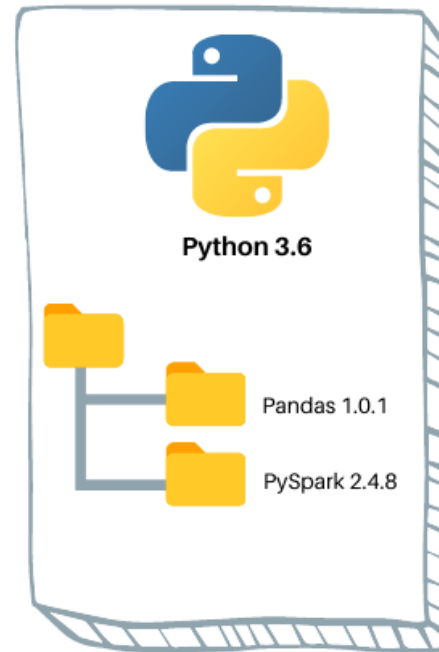


❖ About virtual environment

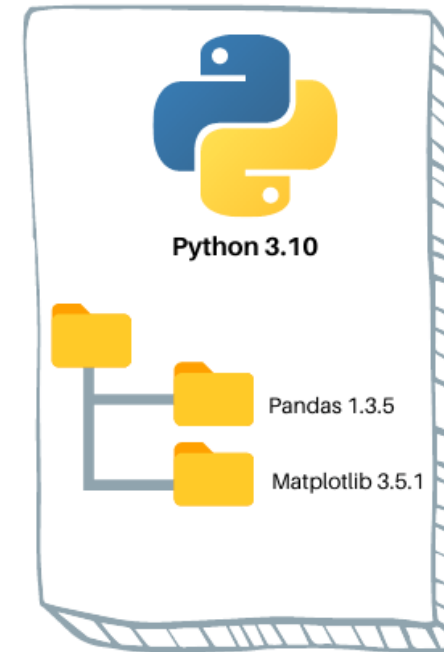
Virtual Environment 1



Virtual Environment 2



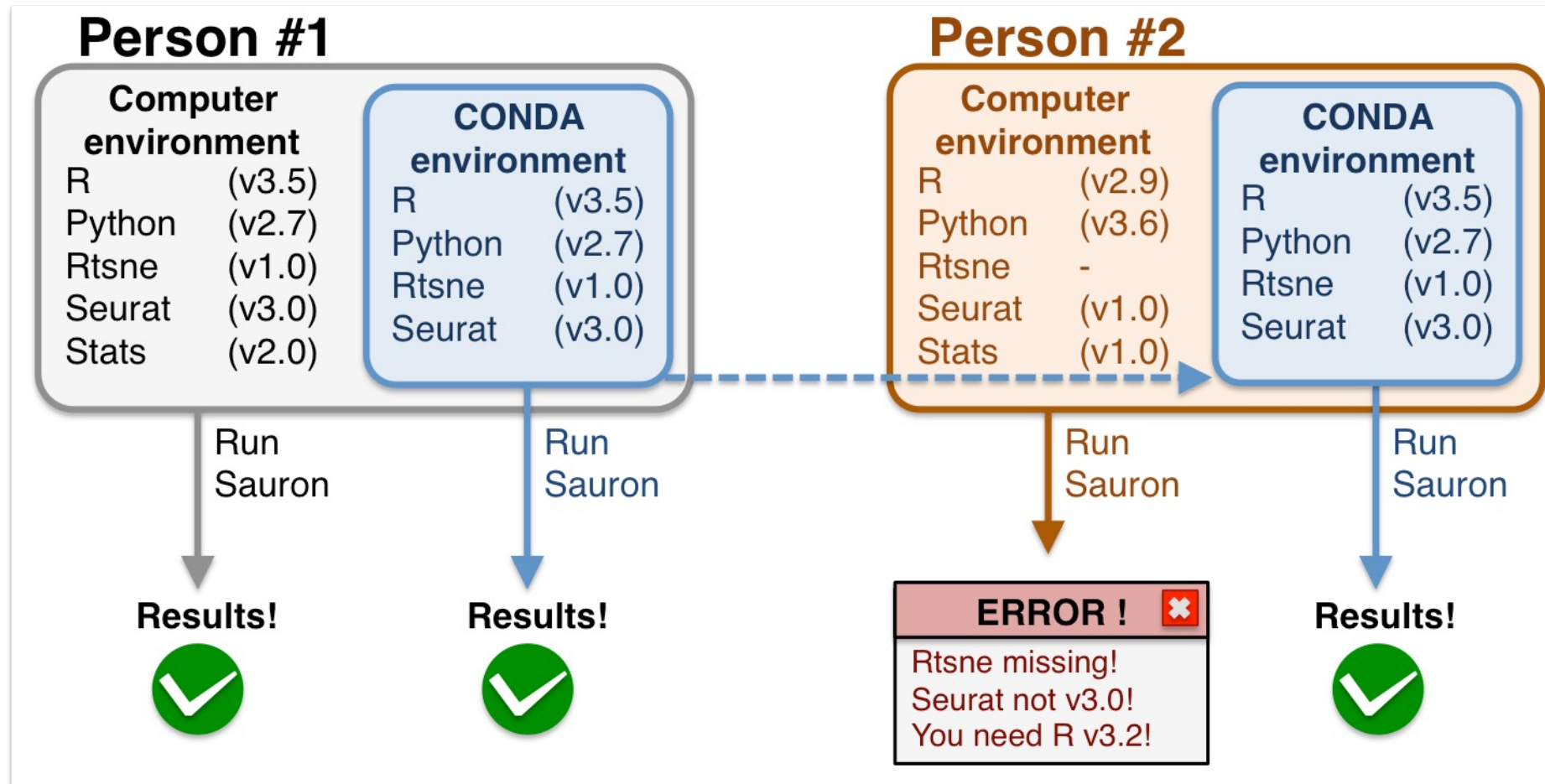
Virtual Environment 3



dataquest.io

labelling

❖ About conda



❖ About conda: Different requirements

yolov10 / requirements.txt

jameslahm update requirements.txt

Code Blame 15 lines (15 loc) · 266 Bytes

```
1 torch==2.0.1
2 torchvision==0.15.2
3 onnx==1.14.0
4 onnxruntime==1.15.1
5 pycocotools==2.0.7
6 PyYAML==6.0.1
7 scipy==1.13.0
8 onnxsim==0.4.36
9 onnxruntime-gpu==1.18.0
10 gradio==4.31.5
11 opencv-python==4.9.0.80
12 psutil==5.9.8
13 py-cpuinfo==9.0.0
14 huggingface-hub==0.23.2
15 safetensors==0.4.3
```

yolov9 / requirements.txt

WongKinYiu Add files via upload

Code Blame 47 lines (41 loc) · 1.05 KB Code 55% faster with GitHub Copilot

```
1 # requirements
2 # Usage: pip install -r requirements.txt
3
4 # Base -----
5 gitpython
6 ipython
7 matplotlib>=3.2.2
8 numpy>=1.18.5
9 opencv-python>=4.1.1
10 Pillow>=7.1.2
11 psutil
12 PyYAML>=5.3.1
13 requests>=2.23.0
14 scipy>=1.4.1
15 thop>=0.1.1
16 torch>=1.7.0
17 torchvision>=0.8.1
18 tqdm>=4.64.0
19 # protobuf<=3.20.1
```

❖ labellmg Installation

Step 3: Create new conda environment

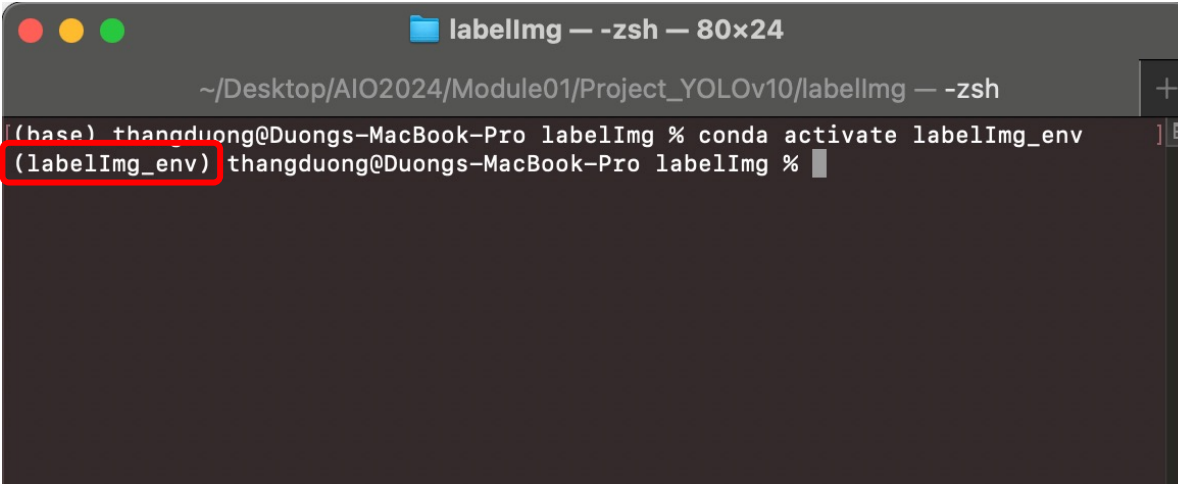
```
[(base) thangduong@Duongs-MacBook-Pro labellmg % conda create -n labellmg_env python=3.11 -y]
Retrieving notices: ...working... done
Collecting package metadata (current_repodata.json): done
Solving environment: done
```

```
Downloading and Extracting Packages

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate labellmg_env
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) thangduong@Duongs-MacBook-Pro labellmg %
```

Step 4: Activate the environment



```
labellmg — zsh — 80x24
~/Desktop/AIO2024/Module01/Project_YOLOv10/labellmg — zsh
[(base) thangduong@Duongs-MacBook-Pro labellmg % conda activate labellmg_env]
(labellmg_env) thangduong@Duongs-MacBook-Pro labellmg %
```

Name of our new environment (**labellmg_env**) is now in the bracket

labellmg

❖ labellmg Installation

Step 5, 6: Install required packages for labellmg

Windows + Anaconda

Download and install [Anaconda](#) (Python 3+)

Open the Anaconda Prompt and go to the [labellmg](#) directory

```
conda install pyqt=5
conda install -c anaconda lxml
pyrcc5 -o libs/resources.py resources.qrc
python labelImg.py
python labelImg.py [IMAGE_PATH] [PRE-DEFINED CLASS FILE]
```

Different system might require additional steps

macOS

Python 3 + Qt5

```
brew install qt # Install qt-5.x.x by Homebrew
brew install libxml2
```

or using pip

```
pip3 install pyqt5 lxml # Install qt and lxml by pip

make qt5py3
python3 labelImg.py
python3 labelImg.py [IMAGE_PATH] [PRE-DEFINED CLASS FILE]
```

Ubuntu Linux

Python 3 + Qt5

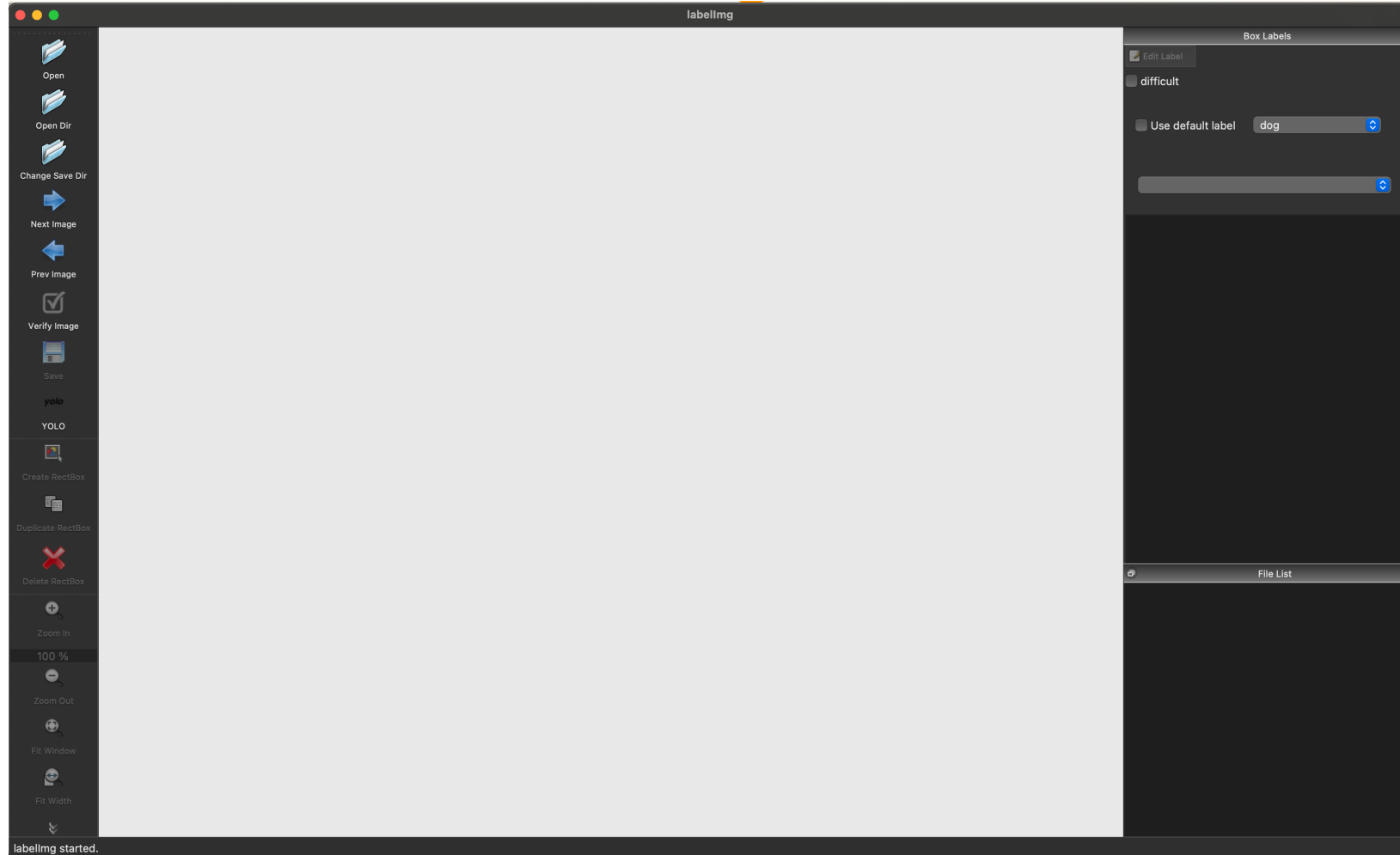
```
sudo apt-get install pyqt5-dev-tools
sudo pip3 install -r requirements/requirements-linux-python3.txt
make qt5py3
python3 labelImg.py
python3 labelImg.py [IMAGE_PATH] [PRE-DEFINED CLASS FILE]
```


❖ labellmg Installation

Step 7: Run labellmg

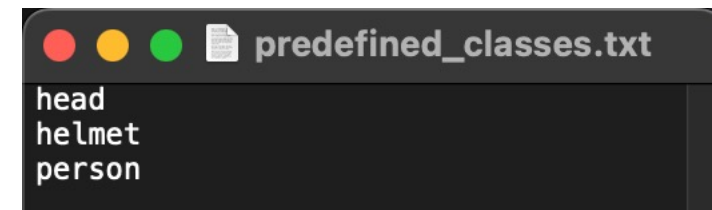
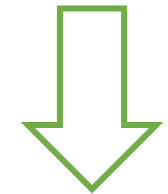
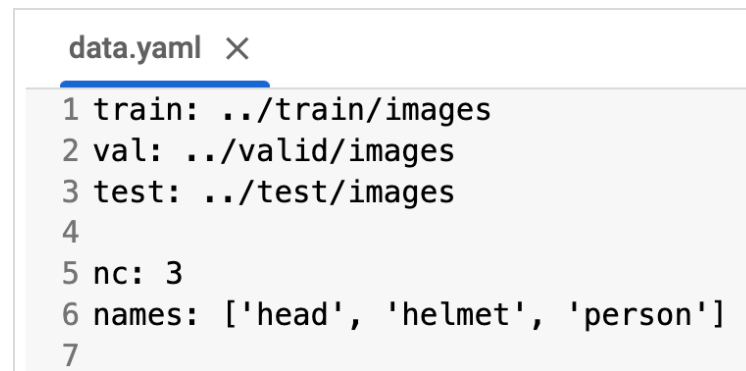
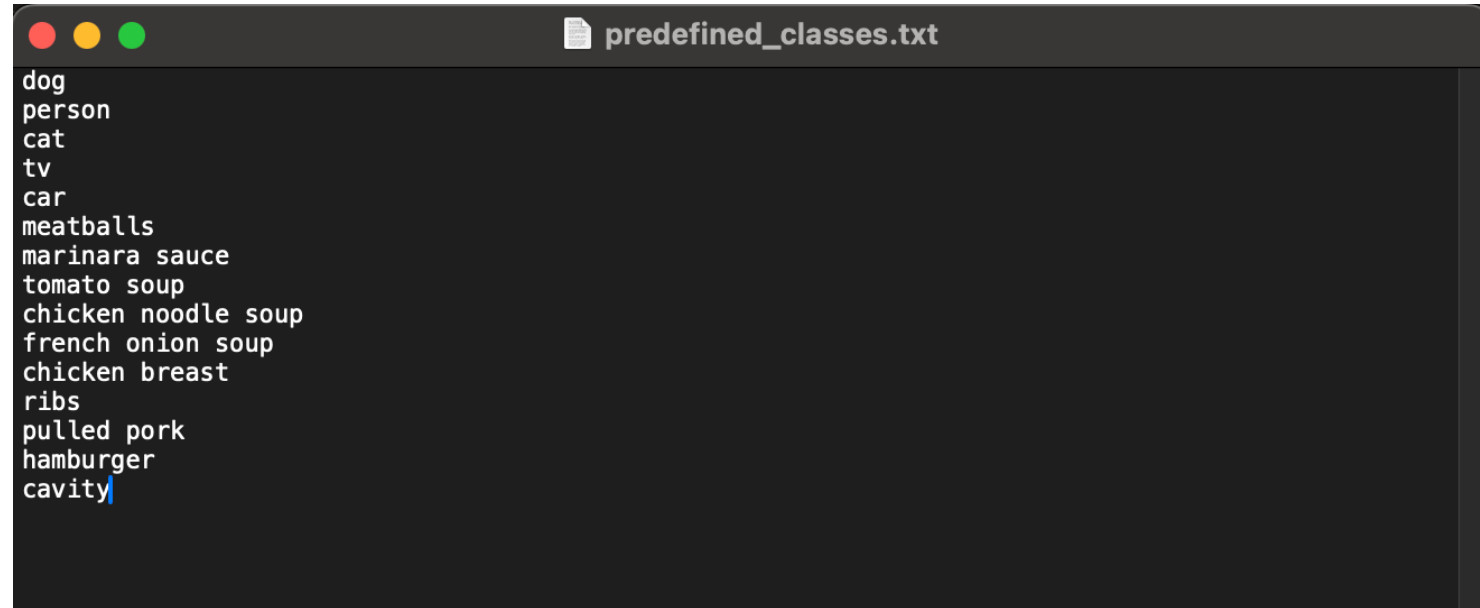
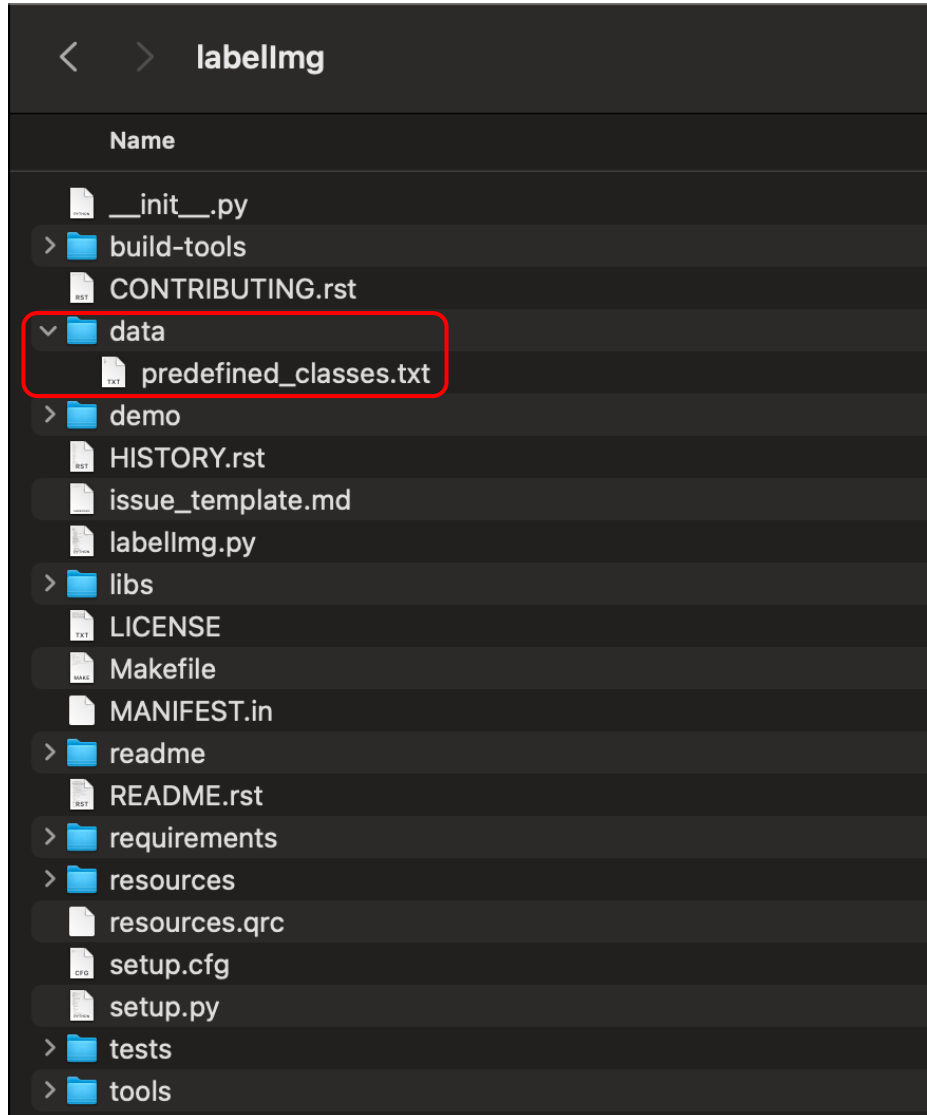
Run this command to open labellmg app:

```
$ python labelImg.py
```



labellmg

❖ How to use?



Change to classnames we work with

labellmg

❖ How to use?

The screenshot shows the labellmg application interface. On the left is a vertical toolbar with various icons. Three red boxes highlight specific buttons: 'Open Dir', 'Change Save Dir', and 'PascalVOC'. Red arrows point from these boxes to three numbered instructions in the center: '1. Open "image-only" folder', '2. Open folder for storing label files', and '3. Change to YOLO'. On the right, a 'Box Labels' panel is visible, showing a list of labels with 'difficult' selected and a dropdown menu set to 'dog'.

1. Open "image-only" folder

2. Open folder for storing label files

3. Change to YOLO

labellmg

❖ How to use?

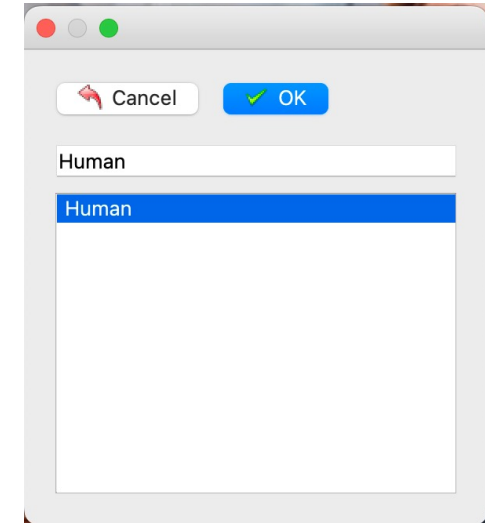
1. Click
"RectBox"



2. Draw the rectangle so that it fits the
object



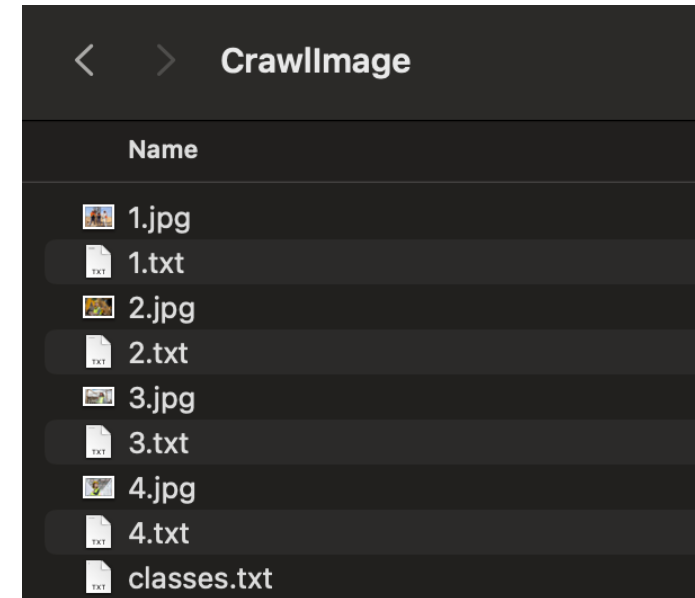
3. Choose the
right class's
name for the
object



Iterate the
process with
other objects

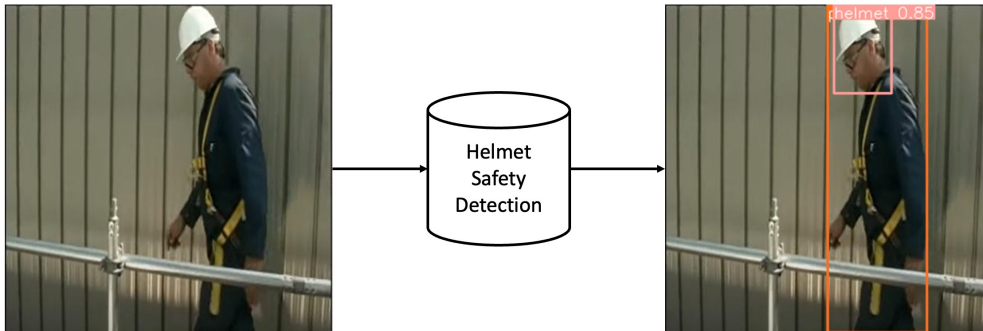


❖ How to use?



Summarization

❖ What we have learned so far



Summarization:

- Object Detection problem in AI (Computer Vision).
Input/Output, Applications...
- Introduce YOLOv10 – a famous architecture with outstanding performance on OD task.
- Learn how to use and train a YOLOv10 model on Helmet Safety Dataset. Also discuss about how ML models are trained.
- Learn about definition and some of the Linux commands.
- Learn how to label new data for Object Detection with labellmg.
- Learn about virtual environment and create it with Conda.

Question

